Theoretical Physics in a Coconutshell

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0.1 Basic Introduction

Quantum field theory (QFT) can be understood as quantum mechanics under special relativity. This book will demonstrate its construction methods and fundamental principles, ensuring that every formula has a derivation process to explain its principles in a simple and clear manner.

Beyond standard QFT, string theory replaces point-like elementary particles with one-dimensional vibrating strings as the fundamental entities, unifying gauge interactions and quantum gravity naturally. Superstring theory incorporates supersymmetry, pairing bosons and fermions to eliminate quantum divergences and support consistent quantum gravity descriptions. M-theory, a unifying framework in 11 dimensions, connects all consistent superstring theories via dualities, regarded as a candidate for a complete theory of everything.

Condensed matter physics applies QFT to many-body systems of matter, studying emergent phenomena like superconductivity, superfluidity and topological order. It focuses on collective excitations, symmetry breaking and topological properties, bridging fundamental quantum principles to real-world material behaviors and quantum functional systems.

0.2 Foundations of Quantum Mechanics

Let us review the derivation of the **Schrödinger equation**: For the wave function $\psi = Ae^{-i(k\mathbf{r}+\omega t)}$, according to the de Broglie relations

$$\boxed{p = \hbar k = \frac{h}{\lambda}, \quad E = \hbar\omega = h\nu}$$

where ω is the angular frequency, k is the wave vector, λ is the wavelength, ν is the frequency, and $\hbar = \frac{h}{2\pi}$ is the reduced Planck constant.

We have

$$\psi = Ae^{-\frac{i}{\hbar}(\mathbf{p}\cdot\mathbf{r}+Et)}$$

Consider the three-dimensional case.

Take the first and second order partial derivatives with respect to t and $\mathbf{r}$:

$$\frac{\partial\psi(\mathbf{r},t)}{\partial t} = -\frac{iE}{\hbar}Ae^{-\frac{i}{\hbar}(Et-\mathbf{p}\cdot\mathbf{r})} = -\frac{iE}{\hbar}\psi(\mathbf{r},t),$$

$$\nabla^2\psi(\mathbf{r},t) = -\frac{\mathbf{p}^2}{\hbar^2}Ae^{-\frac{i}{\hbar}(Et-\mathbf{p}\cdot\mathbf{r})} = -\frac{\mathbf{p}^2}{\hbar^2}\psi(\mathbf{r},t),$$

yielding

$$i\hbar\frac{\partial\psi(\mathbf{r},t)}{\partial t} = E\psi(\mathbf{r},t)$$

According to the classical energy-momentum relation:

$$E = \frac{\mathbf{p}^2}{2m}$$

Considering the Hamiltonian $H = E + V$, we get

$$i\hbar\frac{\partial\psi(\mathbf{r},t)}{\partial t} = H\psi(\mathbf{r},t) = \left(\frac{\mathbf{p}^2}{2m} + V\right)\psi(\mathbf{r},t)$$

Also

$$\nabla^2\psi(\mathbf{r},t) = -\frac{\mathbf{p}^2}{\hbar^2}Ae^{-\frac{i}{\hbar}(Et-\mathbf{p}\cdot\mathbf{r})} = -\frac{\mathbf{p}^2}{\hbar^2}\psi(\mathbf{r},t)$$

That is

$$\mathbf{p} = -i\hbar\nabla$$

Thus we obtain the Schrödinger equation:

$$i\hbar\frac{\partial\psi(\mathbf{r},t)}{\partial t} = \left(-\frac{\hbar^2}{2m}\nabla^2 + V\right)\psi(\mathbf{r},t)$$

Next, we will derive a dynamical equation in QFT based on this method.

0.3 Relativistic Energy-Momentum Relation

We know that the derivation of the Schrödinger equation is based on the classical energy-momentum relation $E = \frac{\mathbf{p}^2}{2m}$, whose drawback is obvious: when electrons move at speeds close to the speed of light, the Schrödinger equation fails. Special relativity was established in 1905. When Schrödinger originally constructed the wave equation for matter waves, he used the energy-momentum relation from special relativity. However, the equation he wrote could not yield the hydrogen atom energy level formula, indicating that equation was wrong.

According to the relativistic energy-momentum relation $E^2 = m^2c^4 + \mathbf{p}^2c^2$, from

$$\frac{\partial^2\psi(\mathbf{r},t)}{\partial t^2} = -\frac{E^2}{\hbar^2}Ae^{-\frac{i}{\hbar}(Et-\mathbf{p}\cdot\mathbf{r})} = -\frac{E^2}{\hbar^2}\psi(\mathbf{r},t), \quad \text{that is} \quad E^2\psi(\mathbf{r},t) = -\hbar^2\frac{\partial^2\psi(\mathbf{r},t)}{\partial t^2}$$

Therefore,

$$E^2\psi = m^2c^4\psi + \mathbf{p}^2c^2\psi$$

$$-\hbar^2\frac{\partial^2\psi}{\partial t^2} = m^2c^4\psi - \hbar^2c^2\nabla^2\psi$$

Dividing both sides by $-\hbar^2c^2$ gives

$$\frac{1}{c^2}\frac{\partial^2\psi}{\partial t^2} = -\frac{m^2c^2}{\hbar^2}\psi + \nabla^2\psi$$

Introducing the d'Alembertian operator:

$$\Box = \frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2$$

Rearranging yields

$$\boxed{\left(\Box + \frac{m^2c^2}{\hbar^2}\right)\psi = 0}$$

The above equation is the so-called Klein-Gordon equation. If we solve this equation, not only does it fail to give the hydrogen atom energy levels, but it also yields negative probabilities and negative energies. In fact, it can only describe spin-0 particles (Higgs bosons, mesons).

0.4 Dirac Equation

Dirac then stepped in. He found that the problem was not simple; the issue lay in the squared energy $E^2\psi$. But if we remove the square, we get a square root that is inconvenient for calculations.

Recalling the perfect square formula we learned in elementary school, $(x+y)^2 = x^2+2xy+y^2$, we naturally ask: can we have $(Ax+By)^2 = x^2+y^2$? Clearly, to satisfy $AB = 0$ and $A^2 = B^2 = 1$, there is no solution for A and B.

But if we change our perspective: $AB \neq BA$, i.e., they do not commute. The answer is almost obvious: these are matrices in linear algebra. Next, we can start deriving the field equation. From $\frac{E^2}{c^2} = m^2c^2 + \mathbf{p}^2$, we get $\frac{E}{c}\psi = mcB\psi + A\mathbf{p}\psi$.

From $\nabla\psi(\mathbf{r},t) = \frac{i\mathbf{p}}{\hbar}Ae^{-\frac{i}{\hbar}(Et-\mathbf{p}\cdot\mathbf{r})} = \frac{i\mathbf{p}}{\hbar}\psi(\mathbf{r},t)$, $\mathbf{p} = -i\hbar\nabla$, obviously we have

$$\tfrac{1}{c}\tfrac{\partial\psi}{\partial t} = -i\hbar\nabla A\psi + mcB\psi$$

yielding the Dirac equation:

$$\boxed{\left(\frac{1}{c}B\frac{\partial}{\partial t} + i\hbar\nabla BA - mc\right)\psi = 0}$$

0.5 Some Notations in Quantum Field Theory

Introduce the Dirac γ matrices, denoted as $\gamma^0 = B, \gamma^1 = BA^1, \gamma^2 = BA^2, \gamma^3 = BA^3$ (note these are superscripts, not exponents; $A = (A^1, A^2, A^3)$). Below we introduce two representations.

0.5.1 Chiral Representation

The core of the chiral representation is to diagonalize γ^5, directly separating left-handed (L) and right-handed (R) chiral components. It is the most commonly used representation in quantum field theory (especially in QCD). Its explicit expressions are as follows:

γ^0	$\begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}$
γ^1	$\begin{pmatrix} 0 & -\sigma_1 \\ \sigma_1 & 0 \end{pmatrix}$
γ^2	$\begin{pmatrix} 0 & -\sigma_2 \\ \sigma_2 & 0 \end{pmatrix}$
γ^3	$\begin{pmatrix} 0 & -\sigma_3 \\ \sigma_3 & 0 \end{pmatrix}$
γ^5	$\begin{pmatrix} -I_2 & 0 \\ 0 & I_2 \end{pmatrix}$

Figure 1: γ matrices in the chiral representation

0.5.2 Dirac Representation (Standard Representation)

The core of the Dirac representation is to diagonalize γ^0, making the "upper part" of the Dirac spinor correspond to the "particle component" in the non-relativistic limit, and the "lower part" correspond to the "antiparticle component". It is suitable for analyzing non-relativistic approximations (e.g., electrons in atomic physics). Its explicit expressions are as follows:

γ^0	$\begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix}$
γ^1	$\begin{pmatrix} 0 & \sigma_1 \\ -\sigma_1 & 0 \end{pmatrix}$
γ^2	$\begin{pmatrix} 0 & \sigma_2 \\ -\sigma_2 & 0 \end{pmatrix}$
γ^3	$\begin{pmatrix} 0 & \sigma_3 \\ -\sigma_3 & 0 \end{pmatrix}$
γ^5	$\begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}$

Figure 2: γ matrices in the Dirac representation

0.5.3 Metric

Regardless of the representation, the γ matrices must satisfy the following **fundamental anticommutation relations** (definition):

$$\{\gamma^\mu, \gamma^\nu\} = \gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2g^{\mu\nu} I_4$$

where:

- $\mu, \nu \in \{0, 1, 2, 3\}$, corresponding to the 4 coordinates of Minkowski spacetime ($x^0 = ct, x^1 = x, x^2 = y, x^3 = z$);
- $g^{\mu\nu}$ is the **Minkowski metric**, i.e., $g^{\mu\nu} = \mathrm{diag}(1, -1, -1, -1)$, the mainstream choice in particle physics;
- I_4 is the 4Œ4 identity matrix.

Additionally, the fifth γ matrix (γ^5) is usually defined to describe chirality (left-/right-handed particles):

$$\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$$

γ^5 anticommutes with all γ^μ ($\mu = 0, 1, 2, 3$), i.e., $\{\gamma^5, \gamma^\mu\} = 0$.

0.5.4 Four-Dimensional Coordinates

To simplify the two equations we derived earlier, we define

$$x^\mu = (x^0, x^1, x^2, x^3) = (x^0, \mathbf{x}) = (t, x, y, z) = \begin{pmatrix} t \\ x \\ y \\ z \end{pmatrix}$$

where $\mathbf{x} = (x^1, x^2, x^3)$, $\mu = 0, 1, 2, 3$, and $x_\mu = (x^0, -x^1, -x^2, -x^3) = (x^0, -\mathbf{x})$. Define the inner product:

$$x^2 = (x_0)^2 - (x_1)^2 - (x_2)^2 - (x_3)^2 = \sum_{\mu,\nu=0}^{3} g_{\mu\nu} x^\mu x^\nu = g_{\mu\nu} x^\mu x^\nu = x^\mu x_\mu$$

using the **Einstein summation convention**: summation symbols are omitted; repeated indices imply summation.

Define

$$\partial_\mu = \frac{\partial}{\partial x^\mu} = \begin{cases} \partial_0 = \frac{\partial}{\partial x^0} = \frac{1}{c}\frac{\partial}{\partial t} & (\mu = 0, \text{time component, describing time rate of change}) \\ \partial_1 = \frac{\partial}{\partial x^1} = \frac{\partial}{\partial x} & (\mu = 1, \text{space-x component, describing spatial change rate in x-direction}) \\ \partial_2 = \frac{\partial}{\partial x^2} = \frac{\partial}{\partial y} & (\mu = 2, \text{space-y component, describing spatial change rate in y-direction}) \\ \partial_3 = \frac{\partial}{\partial x^3} = \frac{\partial}{\partial z} & (\mu = 3, \text{space-z component, describing spatial change rate in z-direction}) \end{cases}$$

$$\partial^\mu = \begin{cases} \partial^0 = g^{00}\partial_0 = \partial_0 = \frac{1}{c}\frac{\partial}{\partial t} & (\text{time component, same as } \partial_0, \text{ because } g^{00} = 1) \\ \partial^1 = g^{11}\partial_1 = -\partial_1 = -\frac{\partial}{\partial x} & (\text{space-x component, sign flipped, because } g^{11} = -1) \\ \partial^2 = g^{22}\partial_2 = -\partial_2 = -\frac{\partial}{\partial y} & (\text{space-y component, sign flipped, because } g^{22} = -1) \\ \partial^3 = g^{33}\partial_3 = -\partial_3 = -\frac{\partial}{\partial z} & (\text{space-z component, sign flipped, because } g^{33} = -1) \end{cases}$$

and $\partial^\mu = g^{\mu\nu}\partial_\nu$.

We have $\Box = \partial^\mu \partial_\mu = \dfrac{1}{c^2}\dfrac{\partial^2}{\partial t^2} - \nabla^2$.

Momentum $p^\mu = \left(\frac{E}{c}, \mathbf{p}\right) = \left(\frac{E}{c}, p_x, p_y, p_z\right) = i\partial^\mu$.

0.5.5 Natural Units

Set

$$\boxed{\hbar = c = 1}$$

Thus, we can simplify the Klein-Gordon equation:

$$\boxed{(\Box + m^2)\psi = 0}$$

Similarly, the Dirac equation:

$$\boxed{(i\hbar\gamma^\mu\partial_\mu - mc)\psi = 0}$$

$$\boxed{(i\gamma^\mu\partial_\mu - m)\psi = 0}$$

Very concise.

$\iint_S \vec{E} \cdot d\vec{S} = \frac{1}{\varepsilon_0} \iiint_V \rho dV$	$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$
$\iint_S \vec{B} \cdot d\vec{S} = 0$	$\nabla \cdot \vec{B} = 0$
$\oint_L \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \iint_S \vec{B} \cdot d\vec{S}$	$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$
$\oint_L \vec{B} \cdot d\vec{l} = \mu_0 \left(\iint_S \vec{J} \cdot d\vec{S} + \varepsilon_0 \frac{d}{dt} \iint_S \vec{E} \cdot d\vec{S} \right)$	$\nabla \times \vec{B} = \mu_0 \left(\vec{J} + \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \right)$

Figure 3: Three-dimensional Maxwell equations

0.6 Starting from Electromagnetism

Note: the unit system used in this section is $\hbar = c = \frac{1}{4\pi\varepsilon_0}$, hence $\mu_0 = 4\pi$.

We know the three-dimensional Maxwell equations:

Below we derive the equations in four dimensions based on Maxwell's equations.

According to the formula in vector analysis: $\nabla \cdot (\nabla \times \mathbf{A}) = 0$.

From $\nabla \cdot \mathbf{B} = 0$, we have:

$$\mathbf{B} = \nabla \times \mathbf{A}$$

Substituting into the third equation, we get $\nabla \times \mathbf{E} = -\nabla \times \frac{\partial \mathbf{A}}{\partial t}$, i.e., $\nabla \times \left(\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t}\right) = 0$. From $\nabla \times \nabla\varphi = 0$, we have $\mathbf{E} = -\nabla\varphi - \frac{\partial \mathbf{A}}{\partial t}$. Substituting $\mathbf{B}$ and $\mathbf{E}$ into the remaining equations yields

$$\frac{\partial^2 \varphi}{\partial t^2} - \nabla^2 \varphi - \frac{\partial}{\partial t}\left(\frac{\partial \varphi}{\partial t} + \nabla \cdot \mathbf{A}\right) = 4\pi\rho$$

$$\frac{\partial^2 \mathbf{A}}{\partial t^2} - \nabla^2 \mathbf{A} + \nabla\left(\frac{\partial \varphi}{\partial t} + \nabla \cdot \mathbf{A}\right) = 4\pi\mathbf{J}$$

We can now elevate this equation to four dimensions. Define $A^\mu = (\varphi, \mathbf{A})$, $J^\mu = (\rho, \mathbf{J})$, obtaining $\Box A^\mu - \partial^\mu(\partial_\nu A^\nu) = 4\pi J^\mu$.

Further simplifying: $\partial_\nu \partial^\nu A^\mu - \partial^\mu \partial_\nu A^\nu = 4\pi J^\mu$, $\partial_\nu(\partial^\nu A^\mu - \partial^\mu A^\nu) = 4\pi J^\mu$. Define $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$, so $\partial_\nu F^{\nu\mu} = 4\pi J^\mu$. Finally, we obtain the electromagnetic field equation

$$\boxed{\partial_\mu F^{\mu\nu} = 4\pi J^\nu}$$

0.7 Lagrangian Density

We know that in analytical mechanics, there is the Euler-Lagrange equation. We can change the coordinate q to ϕ, obtaining

$$\frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right) = 0$$

where the Lagrangian density is $\mathcal{L} = \mathcal{L}(\phi, \partial_\mu\phi)$, and the action is $S = \int d^4x\mathcal{L}(\phi, \partial_\mu\phi)$.

Surely everyone has heard of the Lagrangian of the Standard Model, i.e.,

$$\begin{gathered}
\mathcal{L}_{SM} = -\tfrac{1}{2}\partial_\nu g^a_\mu \partial_\nu g^a_\mu - g_s f^{abc}\partial_\mu g^a_\nu g^b_\mu g^c_\nu - \tfrac{1}{4}g_s^2 f^{abc}f^{ade}g^b_\mu g^c_\nu g^d_\mu g^e_\nu - \partial_\nu W^+_\mu \partial_\nu W^-_\mu - \\
M^2 W^+_\mu W^-_\mu - \tfrac{1}{2}\partial_\nu Z^0_\mu \partial_\nu Z^0_\mu - \tfrac{1}{2c_w^2}M^2 Z^0_\mu Z^0_\mu - \tfrac{1}{2}\partial_\mu A_\nu \partial_\mu A_\nu - igc_w(\partial_\nu Z^0_\mu(W^+_\mu W^-_\nu - \\
W^+_\nu W^-_\mu) - Z^0_\nu(W^+_\mu \partial_\nu W^-_\mu - W^-_\mu \partial_\nu W^+_\mu) + Z^0_\mu(W^+_\nu \partial_\nu W^-_\mu - W^-_\nu \partial_\nu W^+_\mu)) - \\
igs_w(\partial_\nu A_\mu(W^+_\mu W^-_\nu - W^+_\nu W^-_\mu) - A_\nu(W^+_\mu \partial_\nu W^-_\mu - W^-_\mu \partial_\nu W^+_\mu) + A_\mu(W^+_\nu \partial_\nu W^-_\mu - \\
W^-_\nu \partial_\nu W^+_\mu)) - \tfrac{1}{2}g^2 W^+_\mu W^-_\mu W^+_\nu W^-_\nu + \tfrac{1}{2}g^2 W^+_\mu W^-_\nu W^+_\mu W^-_\nu + g^2 c_w^2(Z^0_\mu W^+_\mu Z^0_\nu W^-_\nu - \\
Z^0_\mu Z^0_\mu W^+_\nu W^-_\nu) + g^2 s_w^2(A_\mu W^+_\mu A_\nu W^-_\nu - A_\mu A_\mu W^+_\nu W^-_\nu) + g^2 s_w c_w(A_\mu Z^0_\nu(W^+_\mu W^-_\nu - \\
W^+_\nu W^-_\mu) - 2A_\mu Z^0_\mu W^+_\nu W^-_\nu) - \tfrac{1}{2}\partial_\mu H \partial_\mu H - 2M^2\alpha_h H^2 - \partial_\mu\phi^+\partial_\mu\phi^- - \tfrac{1}{2}\partial_\mu\phi^0\partial_\mu\phi^0 - \\
\beta_h\left(\tfrac{2M^2}{g^2} + \tfrac{2M}{g}H + \tfrac{1}{2}(H^2 + \phi^0\phi^0 + 2\phi^+\phi^-)\right) + \tfrac{2M^4}{g^2}\alpha_h - \\
g\alpha_h M\left(H^3 + H\phi^0\phi^0 + 2H\phi^+\phi^-\right) - \\
\tfrac{1}{8}g^2\alpha_h\left(H^4 + (\phi^0)^4 + 4(\phi^+\phi^-)^2 + 4(\phi^0)^2\phi^+\phi^- + 4H^2\phi^+\phi^- + 2(\phi^0)^2H^2\right) - \\
gMW^+_\mu W^-_\mu H - \tfrac{1}{2}g\tfrac{M}{c_w^2}Z^0_\mu Z^0_\mu H - \\
\tfrac{1}{2}ig\left(W^+_\mu(\phi^0\partial_\mu\phi^- - \phi^-\partial_\mu\phi^0) - W^-_\mu(\phi^0\partial_\mu\phi^+ - \phi^+\partial_\mu\phi^0)\right) + \\
\tfrac{1}{2}g\left(W^+_\mu(H\partial_\mu\phi^- - \phi^-\partial_\mu H) + W^-_\mu(H\partial_\mu\phi^+ - \phi^+\partial_\mu H)\right) + \tfrac{1}{2}g\tfrac{1}{c_w}(Z^0_\mu(H\partial_\mu\phi^0 - \phi^0\partial_\mu H) + \\
M\left(\tfrac{1}{c_w}Z^0_\mu\partial_\mu\phi^0 + W^+_\mu\partial_\mu\phi^- + W^-_\mu\partial_\mu\phi^+\right) - ig\tfrac{s_w^2}{c_w}MZ^0_\mu(W^+_\mu\phi^- - W^-_\mu\phi^+) + igs_w MA_\mu(W^+_\mu\phi^- - \\
W^-_\mu\phi^+) - ig\tfrac{1-2c_w^2}{2c_w}Z^0_\mu(\phi^+\partial_\mu\phi^- - \phi^-\partial_\mu\phi^+) + igs_w A_\mu(\phi^+\partial_\mu\phi^- - \phi^-\partial_\mu\phi^+) - \\
\tfrac{1}{4}g^2 W^+_\mu W^-_\mu\left(H^2 + (\phi^0)^2 + 2\phi^+\phi^-\right) - \tfrac{1}{8}g^2\tfrac{1}{c_w^2}Z^0_\mu Z^0_\mu\left(H^2 + (\phi^0)^2 + 2(2s_w^2 - 1)^2\phi^+\phi^-\right) - \\
\tfrac{1}{2}g^2\tfrac{s_w^2}{c_w}Z^0_\mu\phi^0(W^+_\mu\phi^- + W^-_\mu\phi^+) - \tfrac{1}{2}ig^2\tfrac{s_w^2}{c_w}Z^0_\mu H(W^+_\mu\phi^- - W^-_\mu\phi^+) + \tfrac{1}{2}g^2 s_w A_\mu\phi^0(W^+_\mu\phi^- + \\
W^-_\mu\phi^+) + \tfrac{1}{2}ig^2 s_w A_\mu H(W^+_\mu\phi^- - W^-_\mu\phi^+) - g^2\tfrac{s_w}{c_w}(2c_w^2 - 1)Z^0_\mu A_\mu\phi^+\phi^- - \\
g^2 s_w^2 A_\mu A_\mu\phi^+\phi^- + \tfrac{1}{2}ig_s\,\lambda^a_{ij}(\bar{q}^\sigma_i\gamma^\mu q^\sigma_j)g^a_\mu - \bar{e}^\lambda(\gamma\partial + m^\lambda_e)e^\lambda - \bar{\nu}^\lambda(\gamma\partial + m^\lambda_\nu)\nu^\lambda - \bar{u}^\lambda_j(\gamma\partial + \\
m^\lambda_u)u^\lambda_j - \bar{d}^\lambda_j(\gamma\partial + m^\lambda_d)d^\lambda_j + igs_w A_\mu\left(-(\bar{e}^\lambda\gamma^\mu e^\lambda) + \tfrac{2}{3}(\bar{u}^\lambda_j\gamma^\mu u^\lambda_j) - \tfrac{1}{3}(\bar{d}^\lambda_j\gamma^\mu d^\lambda_j)\right) + \\
\tfrac{ig}{4c_w}Z^0_\mu\{(\bar{\nu}^\lambda\gamma^\mu(1+\gamma^5)\nu^\lambda) + (\bar{e}^\lambda\gamma^\mu(4s_w^2 - 1 - \gamma^5)e^\lambda) + (\bar{d}^\lambda_j\gamma^\mu(\tfrac{4}{3}s_w^2 - 1 - \gamma^5)d^\lambda_j) + \\
(\bar{u}^\lambda_j\gamma^\mu(1 - \tfrac{8}{3}s_w^2 + \gamma^5)u^\lambda_j)\} + \tfrac{ig}{2\sqrt{2}}W^+_\mu\left((\bar{\nu}^\lambda\gamma^\mu(1+\gamma^5)U^{lep}{}_{\lambda\kappa}e^\kappa) + (\bar{u}^\lambda_j\gamma^\mu(1+\gamma^5)C_{\lambda\kappa}d^\kappa_j)\right) + \\
\tfrac{ig}{2\sqrt{2}}W^-_\mu\left((\bar{e}^\kappa U^{lep\dagger}{}_{\kappa\lambda}\gamma^\mu(1+\gamma^5)\nu^\lambda) + (\bar{d}^\kappa_j C^\dagger_{\kappa\lambda}\gamma^\mu(1+\gamma^5)u^\lambda_j)\right) + \\
\tfrac{ig}{2M\sqrt{2}}\phi^+\left(-m^\kappa_e(\bar{\nu}^\lambda U^{lep}{}_{\lambda\kappa}(1-\gamma^5)e^\kappa) + m^\lambda_\nu(\bar{\nu}^\lambda U^{lep}{}_{\lambda\kappa}(1+\gamma^5)e^\kappa)\right) + \\
\tfrac{ig}{2M\sqrt{2}}\phi^-\left(m^\lambda_e(\bar{e}^\lambda U^{lep\dagger}{}_{\lambda\kappa}(1+\gamma^5)\nu^\kappa) - m^\kappa_\nu(\bar{e}^\lambda U^{lep\dagger}{}_{\lambda\kappa}(1-\gamma^5)\nu^\kappa)\right) - \tfrac{g}{2}\tfrac{m^\lambda_\nu}{M}H(\bar{\nu}^\lambda\nu^\lambda) - \\
\tfrac{g}{2}\tfrac{m^\lambda_e}{M}H(\bar{e}^\lambda e^\lambda) + \tfrac{ig}{2}\tfrac{m^\lambda_\nu}{M}\phi^0(\bar{\nu}^\lambda\gamma^5\nu^\lambda) - \tfrac{ig}{2}\tfrac{m^\lambda_e}{M}\phi^0(\bar{e}^\lambda\gamma^5 e^\lambda) - \tfrac{1}{4}\,\bar{\nu}_\lambda\,M^R_{\lambda\kappa}\,(1-\gamma_5)\hat{\nu}_\kappa - \\
\tfrac{1}{4}\,\bar{\nu}_\lambda\,M^R_{\lambda\kappa}\,(1-\gamma_5)\hat{\nu}_\kappa + \tfrac{ig}{2M\sqrt{2}}\phi^+\left(-m^\kappa_d(\bar{u}^\lambda_j C_{\lambda\kappa}(1-\gamma^5)d^\kappa_j) + m^\lambda_u(\bar{u}^\lambda_j C_{\lambda\kappa}(1+\gamma^5)d^\kappa_j)\right) + \\
\tfrac{ig}{2M\sqrt{2}}\phi^-\left(m^\lambda_d(\bar{d}^\lambda_j C^\dagger_{\lambda\kappa}(1+\gamma^5)u^\kappa_j) - m^\kappa_u(\bar{d}^\lambda_j C^\dagger_{\lambda\kappa}(1-\gamma^5)u^\kappa_j)\right) - \tfrac{g}{2}\tfrac{m^\lambda_u}{M}H(\bar{u}^\lambda_j u^\lambda_j) - \\
\tfrac{g}{2}\tfrac{m^\lambda_d}{M}H(\bar{d}^\lambda_j d^\lambda_j) + \tfrac{ig}{2}\tfrac{m^\lambda_u}{M}\phi^0(\bar{u}^\lambda_j\gamma^5 u^\lambda_j) - \tfrac{ig}{2}\tfrac{m^\lambda_d}{M}\phi^0(\bar{d}^\lambda_j\gamma^5 d^\lambda_j) + \bar{G}^a\partial^2 G^a + g_s f^{abc}\partial_\mu\bar{G}^a G^b g^c_\mu + \\
\bar{X}^+(\partial^2 - M^2)X^+ + \bar{X}^-(\partial^2 - M^2)X^- + \bar{X}^0(\partial^2 - \tfrac{M^2}{c_w^2})X^0 + \bar{Y}\partial^2 Y + igc_w W^+_\mu(\partial_\mu\bar{X}^0 X^- - \\
\partial_\mu\bar{X}^+ X^0) + igs_w W^+_\mu(\partial_\mu\bar{Y}X^- - \partial_\mu\bar{X}^+ Y) + igc_w W^-_\mu(\partial_\mu\bar{X}^- X^0 - \\
\partial_\mu\bar{X}^0 X^+) + igs_w W^-_\mu(\partial_\mu\bar{X}^- Y - \partial_\mu\bar{Y}X^+) + igc_w Z^0_\mu(\partial_\mu\bar{X}^+ X^+ - \\
\partial_\mu\bar{X}^- X^-) + igs_w A_\mu(\partial_\mu\bar{X}^+ X^+ - \\
\partial_\mu\bar{X}^- X^-) - \tfrac{1}{2}gM\left(\bar{X}^+ X^+ H + \bar{X}^- X^- H + \tfrac{1}{c_w^2}\bar{X}^0 X^0 H\right) + \tfrac{1-2c_w^2}{2c_w}igM\left(\bar{X}^+ X^0\phi^+ - \bar{X}^- X^0\phi^-\right) + \\
\tfrac{1}{2c_w}igM\left(\bar{X}^0 X^-\phi^+ - \bar{X}^0 X^+\phi^-\right) + igMs_w\left(\bar{X}^0 X^-\phi^+ - \bar{X}^0 X^+\phi^-\right) + \\
\tfrac{1}{2}igM\left(\bar{X}^+ X^+\phi^0 - \bar{X}^- X^-\phi^0\right) .
\end{gathered}$$

Figure 4: Lagrangian of the Standard Model

Although this expression is correct, it makes physics seem very complicated. In fact, it can be greatly simplified using tensors. We will explain the Standard Model comprehensively later.

Based on the field equations we derived earlier, we can reverse-derive the Lagrangian density via the Euler-Lagrange equation. For the Klein-Gordon equation, we have the scalar field $\mathcal{L} = \frac{1}{2}(\partial_\mu \phi \partial^\mu \phi) + \frac{1}{2}m^2\phi$. For the Dirac equation, obviously, $\mathcal{L} = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi$. For the electromagnetic field, it is not hard to notice $-\frac{1}{16\pi}F_{\mu\nu}F^{\mu\nu} + J_\mu A^\mu$.

0.8 Noether's Theorem

As is well known, Noether's theorem states that symmetry (invariance under a set of transformations) corresponds to a conserved current. That is, for the transformation

$$\phi_i(x) \to {\phi'}_i(x') = \phi_i(x) + \delta\phi_i(x)$$

and from $\delta(d^4x) = \partial_\mu(\delta x^\mu)d^4x$, we obtain the variation of the action $\delta S = \int_\Omega \delta\mathcal{L}d^4x + \int_\Omega \mathcal{L} \cdot \partial_\mu(\delta x^\mu)d^4x$.

The variation of the Lagrangian density is $\delta\mathcal{L} = \frac{\partial\mathcal{L}}{\partial\phi_i}\delta\phi_i + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_i)}\delta(\partial_\mu\phi_i)$. Since $\delta(\partial_\mu\phi_i) = \partial_\mu(\delta\phi_i) - \partial_\mu(\delta x^\nu)\partial_\nu\phi_i$, we get

$$\delta\mathcal{L} = \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_i)}\delta\phi_i\right) + \left(\frac{\partial\mathcal{L}}{\partial\phi_i} - \partial_\mu\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_i)}\right)\delta\phi_i - \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_i)}\partial_\mu(\delta x^\nu)\partial_\nu\phi_i$$

Finally, the variation of the action is

$$\delta S = \int_\Omega \partial_\mu\left[\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_i)}\bar{\delta}\phi_i + \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_i)}\partial_\nu\phi_i - \delta^\nu_\mu\mathcal{L}\right)\delta x^\nu\right]d^4x = 0$$

Define the **Noether current** (conserved current):

$$J^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_i)}\bar{\delta}\phi_i + \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_i)}\partial_\nu\phi_i - \delta^\nu_\mu\mathcal{L}\right)\delta x^\nu$$

By Gauss's theorem (divergence theorem), the conserved current satisfies the **continuity equation**:

$$\partial_\mu J^\mu = 0$$

Some common conserved quantities are given below.

Continuous Symmetry (Invariance of Physical System)	Conservation Law	Core Physical Quantity
Time translation symmetry (Time flow does not change system laws)	Energy conservation law	Energy (E)
Space translation symmetry (Spatial position does not change system laws)	Momentum conservation law	Momentum (p)
Space rotation symmetry (Spatial orientation does not change system laws)	Angular momentum conservation law	Angular momentum (L)
U(1) gauge symmetry (Phase invariance of electromagnetic interaction)	Charge conservation law	Charge (Q)
SU(2) gauge symmetry (Isospin invariance of weak interaction)	Isospin conservation law	Isospin (I)
SU(3) gauge symmetry (Color charge invariance of strong interaction)	Color charge conservation law	Color charge (R / G / B)

Figure 5: Common symmetries and corresponding conserved quantities

Let's derive them.

0.8.1 Example 1: Time translation symmetry Energy conservation

- Time translation: $\delta x^0 = \epsilon$ (ϵ infinitesimal constant), $\delta x^{1,2,3} = 0$; intrinsic variation $\bar{\delta}\phi_i = 0$ (only spacetime translation, fields themselves unchanged).
- Substituting, take $\mu = 0$ (time component), define **energy density** $\rho = J^0$. Combined with continuity equation $\partial_0 \rho + \nabla \cdot \boldsymbol{J} = 0$ ($\boldsymbol{J}$ spatial components), integrate to get

total energy:

$$E = \int \rho \, d^3x = \int \left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}_i} \dot{\phi}_i - \mathcal{L} \right) d^3x = const$$

i.e., the **law of energy conservation**.

0.8.2 Example 2: Spatial translation symmetry Momentum conservation

- Spatial translation (along x^k direction): $\delta x^k = \epsilon$, $\delta x^0 = \delta x^{l \neq k} = 0$; $\bar{\delta}\phi_i = 0$.
- Substituting, take $\mu = k$ (space component), define **momentum density** $p^k = J^k$, integrate to get total momentum:

$$P^k = \int p^k \, d^3x = \int \frac{\partial \mathcal{L}}{\partial(\partial_k \phi_i)} \partial_0 \phi_i \, d^3x = const$$

i.e., the **law of momentum conservation**.

0.8.3 Example 3: U(1) gauge symmetry Charge conservation

- U(1) gauge transformation (scalar field example): $\phi \to \phi' = e^{iq\alpha}\phi$, infinitesimal transformation $\bar{\delta}\phi = iq\alpha\phi$ (α infinitesimal constant); $\delta x^\mu = 0$.
- Lagrangian density $\mathcal{L} = \partial_\mu \phi^* \partial^\mu \phi - m^2 |\phi|^2$, substitute to get conserved current:

$$J^\mu = iq \left(\phi^* \partial^\mu \phi - \phi \partial^\mu \phi^* \right)$$

Integrate to get total charge:

$$Q = \int J^0 \, d^3x = iq \int \left(\phi^* \dot{\phi} - \phi \dot{\phi}^* \right) d^3x = const$$

i.e., the **law of charge conservation**.

0.8.4 Exercise 1.7.1

Based on the scalar field $\mathcal{L} = \frac{1}{2}(\partial_\mu \phi \partial^\mu \phi) + \frac{1}{2} m^2 \phi$, the spinor field $\mathcal{L} = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi$, and the electromagnetic field $-\frac{1}{16\pi} F_{\mu\nu} F^{\mu\nu} + J_\mu A^\mu$, derive the corresponding equations of motion.

0.8.5 Exercise 1.8.1

Derive other conserved quantities based on Noether's theorem.

1. Path Integral and Feynman Diagrams

Let's formally enter quantum field theory.

1.1 Plane Wave Expansion of Scalar Fields

For the Klein-Gordon equation:

$$\boxed{(\Box + m^2)\psi = 0}$$

We study its solutions. First, assume a solution of the wave function form $\varphi(x) = e^{-ip^\mu \cdot x}$, then $\Box\varphi = -(p^\mu)^2\varphi$. Substituting into the original equation yields $-((p^0)^2 - |\mathbf{p}|^2 - m^2)\varphi = 0$. Clearly, $(p^0)^2 = |\mathbf{p}|^2 + m^2$, giving $p^0 = E_\mathbf{p} = \pm\sqrt{|\mathbf{p}|^2 + m^2}$. Normalized in Fourier integral form, split into positive and negative frequency parts (corresponding to creation and annihilation):

$$\boxed{\phi(x) = \int \frac{d^3p}{(2\pi)^{3/2}} \frac{1}{\sqrt{2E_\mathbf{p}}} \left(\underbrace{a_\mathbf{p} e^{-ip\cdot x}}_{\text{annihilation part}} + \underbrace{a_\mathbf{p}^\dagger e^{ip\cdot x}}_{\text{creation part}} \right)}$$

The above is called the **plane wave expansion of a real scalar field**. Here, $a_\mathbf{p}$ is the annihilation operator, $a_\mathbf{p}^\dagger$ is the creation operator, satisfying the commutation relations $[a_\mathbf{p}, a_\mathbf{q}^\dagger] = (2\pi)^3\delta^{(3)}(\mathbf{p} - \mathbf{q}), \quad [a_\mathbf{p}, a_\mathbf{q}] = [a_\mathbf{p}^\dagger, a_\mathbf{q}^\dagger] = 0$, with $\delta^{(3)}$ the three-dimensional Dirac delta function, $\delta^{(3)}(\mathbf{x} - \mathbf{y}) = \delta(x_1 - y_1)\delta(x_2 - y_2)\delta(x_3 - y_3)$.

1.2 Derivation of Feynman Propagator and Green's Function

The classical Green's function $G(x - y)$ is defined as the solution satisfying:

$$(\Box_x + m^2)G(x - y) = \delta^4(x - y)$$

Solve via Fourier transform:

Perform Fourier transform on the Green's function:

$$G(x-y)=\int \frac{d^4p}{(2\pi)^4}e^{-ip\cdot(x-y)}\tilde{G}(p)$$

Substitute into the field equation, using $\Box e^{-ip\cdot x}=-p^2e^{-ip\cdot x}$, to get:

$$(-p^2+m^2)\tilde{G}(p)=1$$

Thus the Green's function in momentum space is:

$$\tilde{G}(p)=\frac{1}{m^2-p^2}$$

Then inverse Fourier transform yields the Green's function in coordinate space:

$$G(x-y)=\int \frac{d^4p}{(2\pi)^4}\frac{e^{-ip\cdot(x-y)}}{m^2-p^2}$$

Define the step function $\theta(t)=\begin{cases}1 & \text{if } t>0\\ 0 & \text{if } t<0\end{cases}$, obviously $\theta'(x)=\delta(x)$. Define the **time-ordered product** $\mathcal{T}[\phi(x)\phi(y)]=\theta(x^0-y^0)\phi(x)\phi(y)\pm\theta(y^0-x^0)\phi(y)\phi(x)$. In quantum mechanics: $\langle final\ state|H|initial\ state\rangle$ represents the process from initial to final state. Clearly, $\langle 0|H|0\rangle$ represents the vacuum.

Define the **contraction** of scalar fields: $\contraction{}{\phi}{(x)}{\phi}\phi(x)\phi(y)=\overline{\phi(x)\phi(y)}=\langle 0|\mathcal{T}[\phi(x)\phi(y)]|0\rangle=D_F(x-y)$, where the Green's function for scalar fields is $G_F(x-y)=D_F(x-y)=\int\frac{d^4p}{(2\pi)^4}\frac{i}{p^2-m^2+i\epsilon}e^{-ip\cdot(x-y)}$. We naturally ask: what is the use of $i\epsilon$? Imagine when $p^2-m^2=0$, the ordinary Green's function diverges. Then we have $\frac{i}{i\epsilon}=\frac{1}{\epsilon}$, ϵ is an infinitesimal, $i\epsilon$ shifts slightly off the real axis, and $\frac{1}{\epsilon}$ ensures no true infinity.

Other propagators include the Dirac spinor field propagator $S_F(x-y)=\int\frac{d^4p}{(2\pi)^4}\frac{i(\gamma^\mu p_\mu+m)}{p^2-m^2+i\epsilon}e^{-ip\cdot(x-y)}$, and the photon field propagator $D_{F,\mu\nu}(x-y)=\int\frac{d^4p}{(2\pi)^4}\frac{-ig_{\mu\nu}}{p^2+i\epsilon}e^{-ip\cdot(x-y)}$. Sometimes denoted as $\not{p}=\gamma^\mu p_\mu$.

1.2.1 Exercise 2.2.1

Derive the spinor field propagator from the Dirac equation.

1.3 Derivation of Feynman Propagator for Spinor/Vector/Scalar Fields

Definition: $D_F(x-y)=\contraction{}{\phi}{(x)}{\phi}\phi(x)\phi(y)=\langle 0|T[\phi(x)\phi(y)]|0\rangle$

For Feynman propagator, use plane wave expansion of $\phi(x)$:

$$\phi^+(x)=\int\frac{\mathrm{d}^3p}{(2\pi)^3}\frac{1}{2E_p}a_p\mathrm{e}^{-\mathrm{i}p\cdot x},\quad \phi^-(x)=\int\frac{\mathrm{d}^3p}{(2\pi)^3}\frac{1}{2E_p}a_p^\dagger\mathrm{e}^{\mathrm{i}p\cdot x}$$

When $x^0>y^0$, we have:

$$\langle 0|T[\phi(x)\phi(y)]|0\rangle=\langle 0|\phi(x)\phi(y)|0\rangle=\langle 0|\phi^+(x)\phi^-(y)|0\rangle$$

$$=\int\frac{\mathrm{d}^3p\mathrm{d}^3q}{(2\pi)^6 4E_pE_q}\langle 0|a_p\mathrm{e}^{-\mathrm{i}p\cdot x}a_q^\dagger\mathrm{e}^{\mathrm{i}q\cdot y}|0\rangle$$

$$= \int \frac{\mathrm{d}^3p\mathrm{d}^3q}{(2\pi)^6 4E_pE_q}\mathrm{e}^{-\mathrm{i}(p\cdot x-q\cdot y)}\langle 0|[a_p, a_q^\dagger]|0\rangle$$

Using $[a_p, a_q^\dagger] = (2\pi)^3\delta^{(3)}(p-q)$, so the term is zero when $p \neq q$. Thus we require $p = q$, and:

$$D_F(x-y) = \int \frac{\mathrm{d}^3p}{(2\pi)^3}\frac{\mathrm{e}^{\mathrm{i}p\cdot(x-y)}}{2E_p} = \int \frac{\mathrm{d}^3p}{(2\pi)^3}\frac{\mathrm{e}^{\mathrm{i}p\cdot(x-y)}\mathrm{e}^{-\mathrm{i}E_p(x^0-y^0)}}{2E_p}$$

When $x^0 < y^0$, similarly:

$$D_F(x-y) = \int \frac{\mathrm{d}^3p}{(2\pi)^3}\frac{\mathrm{e}^{\mathrm{i}p\cdot(x-y)}}{2E_p}$$

Where the momentum integral variable p is replaced by $-p$.
In summary:

$$D_F(x-y) = \int \frac{\mathrm{d}^3p}{(2\pi)^3}\mathrm{e}^{\mathrm{i}p\cdot(x-y)}\left[\theta(x^0-y^0)\frac{\mathrm{e}^{-\mathrm{i}E_p(x^0-y^0)}}{2E_p} + \theta(y^0-x^0)\frac{\mathrm{e}^{\mathrm{i}E_p(x^0-y^0)}}{2E_p}\right]$$

Note:

$$-\frac{1}{\pi}\int_{-\infty}^{+\infty}\frac{\mathrm{e}^{-\mathrm{i}E_p(x^0-y^0)}}{(E_p-\mathrm{i}\epsilon)(E_p+\mathrm{i}\epsilon)}\mathrm{d}E_p = \left[\theta(x^0-y^0)\frac{\mathrm{e}^{-\mathrm{i}E_p(x^0-y^0)}}{2E_p} + \theta(y^0-x^0)\frac{\mathrm{e}^{\mathrm{i}E_p(x^0-y^0)}}{2E_p}\right]$$

Taking $E_p - \mathrm{i}\epsilon$ and $E_p + \mathrm{i}\epsilon$ as poles, discuss for $x^0 > y^0$ and $x^0 < y^0$ respectively (Residue Theorem, easy to prove). Also:

$$[p^0 - (E_p - \mathrm{i}\epsilon)][p^0 + (E_p - \mathrm{i}\epsilon)] = (p^0)^2 - (E_p - \mathrm{i}\epsilon)^2 = (p^0)^2 - E_p^2 + 2\mathrm{i}\epsilon E_p + \epsilon^2$$

From $E_p^2 = |p|^2 + m^2$:

$$(p^0)^2 - |p|^2 - m^2 + 2\mathrm{i}\epsilon E_p + \epsilon^2 \approx p^2 - m^2 + \mathrm{i}\epsilon$$

Since $2E_p\epsilon \approx \epsilon$ (ignore ϵ^2, take $\epsilon \to 0^+$ finally), we get:

$$D_F(x-y) = \int \frac{\mathrm{d}^4p}{(2\pi)^4}\frac{\mathrm{e}^{\mathrm{i}p\cdot(x-y)}}{p^2 - m^2 + \mathrm{i}\epsilon}$$

1.3.1 Feynman Propagator for Vector Field

$$\Delta_F^{\mu\nu}(x-y) = \overbracket{A^\mu(x)A^\nu}(y) = \langle 0|T[A^\mu(x)A^\nu(y)]|0\rangle$$

From $A^\mu(x,t) = \int \frac{\mathrm{d}^3p}{(2\pi)^3}\frac{1}{\sqrt{2E_p}}\left[\varepsilon^\mu(p,\lambda)a_{p\lambda}\mathrm{e}^{-\mathrm{i}p\cdot x} + \varepsilon^{\mu *}(p,\lambda)a_{p\lambda}^\dagger\mathrm{e}^{\mathrm{i}p\cdot x}\right]$ (similar to scalar field method):

$$\langle 0|A^\mu(x)A^\nu(y)|0\rangle = \int \frac{\mathrm{d}^3p}{(2\pi)^3}\frac{\mathrm{e}^{\mathrm{i}p\cdot(x-y)}}{2E_p}\sum_\lambda \varepsilon^\mu(p,\lambda)\varepsilon^\nu(p,\lambda)$$

From $\sum_\lambda \varepsilon_\mu(p,\lambda)\varepsilon_\nu(p,\lambda) = g_{\mu\nu}$: We have:

$$\sum_\lambda \varepsilon^\mu(p,\lambda)\varepsilon^\nu(p,\lambda) = \sum_{\lambda=1}^{4} \varepsilon^\mu(p,\lambda)\varepsilon^\nu(p,\lambda)$$

$$= -\frac{g^{\mu\nu}}{2}\varepsilon_\mu(p,0)\varepsilon_\nu(p,0) + g^{\mu\nu}\varepsilon_\mu(p,0)\varepsilon_\nu(p,0)$$

$$= -g^{\mu\nu} + \frac{p^\mu p^\nu}{m^2}$$

Called polarization sum relation, organize for $x^0 < y^0, x^0 > y^0$:

$$\Delta_F^{\mu\nu}(x-y) = \int \frac{\mathrm{d}^3 p}{(2\pi)^3} \frac{(-g^{\mu\nu} + p^\mu p^\nu/m^2)}{2E_p} \left[\theta(x^0 - y^0)\mathrm{e}^{-\mathrm{i}p\cdot(x-y)} + \theta(y^0 - x^0)\mathrm{e}^{\mathrm{i}p\cdot(x-y)}\right]$$

Using scalar field method:

$$\Delta_F^{\mu\nu}(x-y) = \int \frac{\mathrm{d}^4 p}{(2\pi)^4} \frac{-(g^{\mu\nu} - p^\mu p^\nu/m^2)}{p^2 - m^2 + \mathrm{i}\epsilon}\mathrm{e}^{\mathrm{i}p\cdot(x-y)}$$

For massless case (photon):

$$\Delta_F^{\mu\nu}(x-y) = \int \frac{\mathrm{d}^4 p}{(2\pi)^4} \frac{-g^{\mu\nu}}{p^2 + \mathrm{i}\epsilon}\mathrm{e}^{\mathrm{i}p\cdot(x-y)}$$

In scalar field:

$$= \langle 0|T[\psi_\alpha(x)\overline{\psi}_\beta(y)]|0\rangle$$

$$\psi_\alpha^{(+)}(x) = \int \frac{\mathrm{d}^3 p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} \sum_s u_\alpha(p,\lambda) b_{ps} \mathrm{e}^{-\mathrm{i}p\cdot x}$$

$$\overline{\psi}_\beta^{(-)}(x) = \int \frac{\mathrm{d}^3 p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} \sum_s \overline{v}_\beta(p,\lambda) d_{ps}^\dagger \mathrm{e}^{\mathrm{i}p\cdot x}$$

Also from anticommutation relation:

$$\sum_s u(p,\lambda)\overline{u}(p,\lambda) = p + m, \quad \sum_s v(p,\lambda)\overline{v}(p,\lambda) = p - m$$

1.4 Feynman Diagrams and Scattering Amplitudes

We can represent via Feynman diagram:

- $x - - - - y = \phi(y)\phi(x) = D_F(y-x)$
- $x \sim\sim\sim y = A^\mu(y)A^\nu(x) = \Delta_F^{\mu\nu}(y-x)$
- $x \longrightarrow y = \psi(y)\overline{\psi}(x) = S_F(y-x)$

Obtain:

$$
\begin{aligned}
S_{F\alpha\beta}(x-y) &= \langle 0|T[\psi_\alpha(x)\overline{\psi}_\beta(y)]|0\rangle \\
&= \int \frac{\mathrm{d}^4 p}{(2\pi)^4}\frac{(p+m)_{\alpha\beta}}{p^2-m^2+\mathrm{i}\epsilon}\mathrm{e}^{\mathrm{i}p\cdot(x-y)} \\
&= \int \frac{\mathrm{d}^4 p}{(2\pi)^4}\frac{(p-m)_{\alpha\beta}}{p^2-m^2+\mathrm{i}\epsilon}\mathrm{e}^{\mathrm{i}p\cdot(x-y)} \\
&= \int \frac{\mathrm{d}^4 p}{(2\pi)^4}\frac{1}{p^2-m^2+\mathrm{i}\epsilon}\mathrm{e}^{\mathrm{i}p\cdot(x-y)}
\end{aligned}
$$

1.5 Feynman Diagrams

Consider the simplest case: $\langle 0|\mathcal{T}[\phi(x_1)\phi(x_2)]|0\rangle = D_F(x_1-x_2)$. We have the following Feynman diagram:

Figure 1.1: Feynman diagram for the two-point function

This doesn't have a clear physical meaning yet; it's purely a man-made theory. Next, consider a more complex case: ϕ^4 theory.

1.5.1 Wick's Theorem

Before that, it's necessary to understand Wick's theorem:

$$
\boxed{\mathcal{T}\{\phi(x_1)\phi(x_2)\cdots\phi(x_n)\} = \mathcal{N}\{\phi(x_1)\phi(x_2)\cdots\phi(x_n)\} + \sum_{\text{all possible contractions}}}
$$

You may not understand what "all possible contractions" means; we will explain below.

The normal product (Normal Product) is defined as: arranging operators in the order "creation operators on the left, annihilation operators on the right", and when swapping operators (add a minus sign for fermion fields, none for boson fields).

- For two-point fields: $\mathcal{N}\{\phi(x)\phi(y)\} = \mathcal{N}\{(\phi^+(x)+\phi^-(x))(\phi^+(y)+\phi^-(y))\} = \phi^-(x)\phi^-(y)+\phi^-(x)\phi^+(y)+\phi^-(y)\phi^+(x)+\phi^+(x)\phi^+(y)$ (no minus sign for boson field exchange);
- Key property: **The vacuum expectation value of a normal product is 0**, i.e., $\langle 0|\mathcal{N}\{\text{any number of field operators}\}|0\rangle = 0$ (because the vacuum $|0\rangle$ is an eigenstate of annihilation operators, $\hat{a}_{\boldsymbol{p}}|0\rangle = 0$, and the normal product always has an annihilation operator on the rightmost, acting on the vacuum gives 0).

1.5.2 ϕ^4 Theory

According to Wick's theorem, we have:

$$
\begin{aligned}
T\{\phi_1\phi_2\phi_3\phi_4\} = N\{&\phi_1\phi_2\phi_3\phi_4 + \phi_1\phi_2\phi_3\phi_4 + \phi_1\phi_2\phi_3\phi_4 + \phi_1\phi_2\phi_3\phi_4 \\
&+ \phi_1\phi_2\phi_3\phi_4 + \phi_1\phi_2\phi_3\phi_4 + \phi_1\phi_2\phi_3\phi_4 \\
&+ \phi_1\phi_2\phi_3\phi_4 + \phi_1\phi_2\phi_3\phi_4 + \phi_1\phi_2\phi_3\phi_4\}.
\end{aligned}
$$

Figure 1.2: Wick contractions for the four-point function

Thus we obtain

$$
\begin{aligned}
&\langle 0|T\{\phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4)\}|0\rangle \\
&= \mathrm{D}_F(x_1 - x_2)D_F(x_3 - x_4) \\
&\quad + \mathrm{D}_F(x_1 - x_3)D_F(x_2 - x_4) \\
&\quad + \mathrm{D}_F(x_1 - x_4)D_F(x_2 - x_3)
\end{aligned}
$$

We can draw its Feynman diagram:

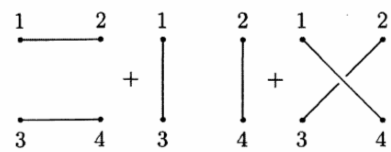
$\langle 0|\, T\{\phi_1\phi_2\phi_3\phi_4\}\, |0\rangle =$ 

Figure 1.3: Feynman diagram for the four-point function

Following this method, try to draw a Feynman diagram for a complex analytic expression:

$$
\begin{aligned}
&\langle 0|\, \phi(x)\phi(y)\, \tfrac{1}{3!}\left(\tfrac{-i\lambda}{4!}\right)^3 \int d^4z\, \phi\phi\phi\phi \int d^4w\, \phi\phi\phi\phi \int d^4u\, \phi\phi\phi\phi\, |0\rangle \\
&= \frac{1}{3!}\left(\frac{-i\lambda}{4!}\right)^3 \int d^4z\, d^4w\, d^4u\; D_F(x-z)D_F(z-z)D_F(z-w) \\
&\qquad \times D_F(w-y)D_F^2(w-u)D_F(u-u).
\end{aligned}
$$

Figure 1.4: Analytic expression for a higher-order loop diagram

Clearly, we have:

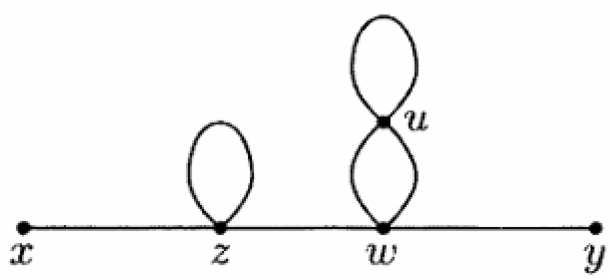


Figure 1.5: Feynman diagram corresponding to the higher-order loop diagram

1.5.3 Feynman Rules

Based on the propagators, we have the following Feynman rules:

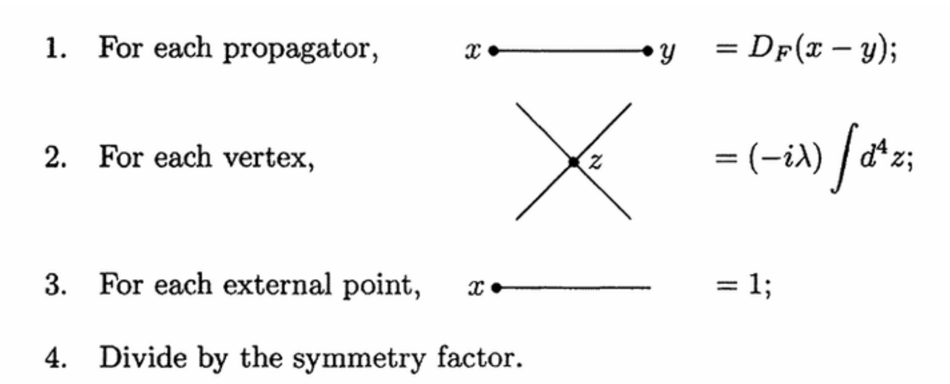


Figure 1.6: Basic symbols for Feynman rules

In different fields, the rules are similar. Scalar field:

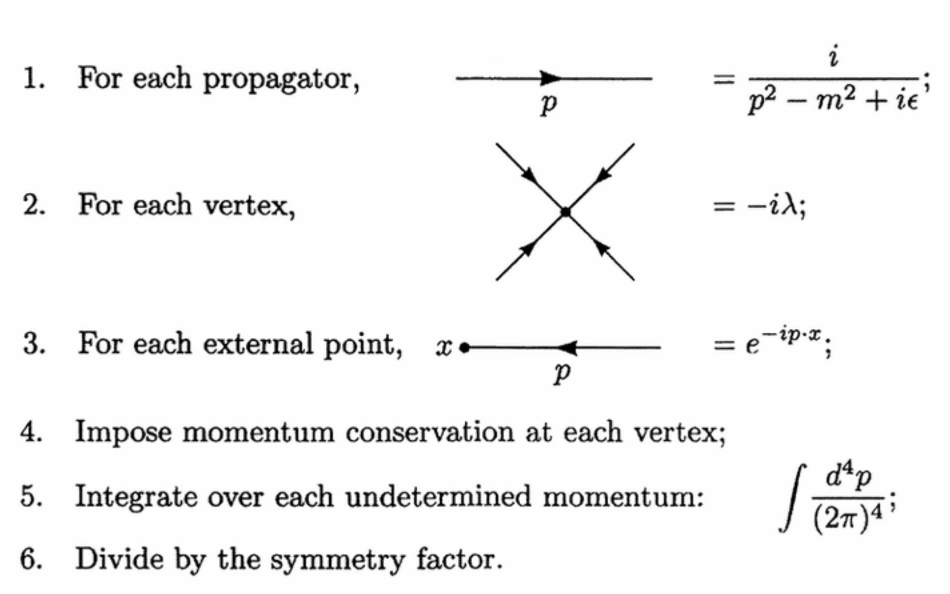


Figure 1.7: Feynman rules for scalar fields

Spinor field:

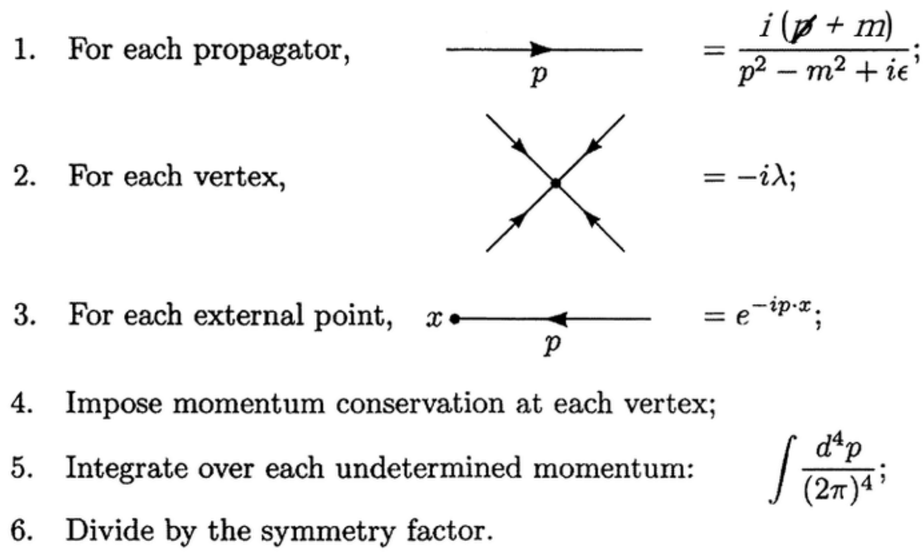


Figure 1.8: Feynman rules for spinor fields

1.5.4 Exercise 2.3.1

Prove Wick's theorem.

1.6 Feynman Path Integral

We know the propagator is defined as $K(x_f, t_f; x_i, t_i) = \langle x_f, t_f | x_i, t_i \rangle$ from one time point to another. Expressing with the time evolution operator U, $K(x_f, t_f; x_i, t_i) = \langle x_f | U(t_f, t_i) | x_i \rangle$. Decompose into many small time intervals: $U(t_f, t_i) = \prod_{k=1}^{N} U(t_k, t_{k-1})$. According to the definition, $U(t_k, t_{k-1}) = e^{-\frac{i\Delta t}{\hbar}H} = exp(-\frac{i\Delta t}{\hbar}H)$. Substituting yields:

$$\boxed{K(x_f, t_f; x_i, t_i) = \int \mathscr{D}x(t) \exp\left(\frac{i}{\hbar} S[x(t)]\right)}$$

called the Feynman path integral. Sometimes written as $Z = \int \mathscr{D}x(t) \exp\left(\frac{i}{\hbar} S[x(t)]\right)$, Z is called the partition function.

1.7 Scattering Theory

1.7.1 Basic Concepts

The scattering amplitude is a complex function describing the probability amplitude for particle collision processes. In quantum mechanics, probability is determined by the modulus square of the probability amplitude (wave function). The scattering amplitude $\mathcal{M}$ is precisely the transition probability amplitude from the incoming particle state to the outgoing particle stateits modulus square $|\mathcal{M}|^2$ is directly proportional to the "probability of scattering (or decay) occurring".

When particles collide (scatter), the scattering amplitude $\mathcal{M}$ cannot be directly measured experimentally, but the distribution probability of outgoing particles in different directions can be measured. This distribution is described by the differential cross section, denoted $\frac{d\sigma}{d\Omega}$ (Ω is the solid angle), where σ is the total cross section.

The differential cross section is the observable manifestation of the scattering amplitude. The two are directly related via Fermi's Golden Rule:

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 s} \cdot \frac{|\vec{p}_f|}{|\vec{p}_i|} \cdot |\mathcal{M}|^2$$

where s is a Mandelstam variable, $s = (2E)^2$.

The decay rate describes the probability of an unstable particle (e.g., neutron, meson, Higgs boson) spontaneously splitting into lighter particles, denoted Γ. The mean lifetime of the particle is $\tau = 1/\Gamma$.

1.7.2 Relevant Formulas

The outgoing state $|f\rangle$ can be expressed as the incoming state $|i\rangle$ acted upon by the S-matrix: $|f\rangle_{\text{out}} = S \cdot |i\rangle_{\text{in}}$. Since transition probability in quantum mechanics is defined by the inner product of state vectors, the matrix element of the S-matrix (denoted $\langle f|S|i\rangle$) directly corresponds to the "probability amplitude for transition from incoming state $|i\rangle$ to outgoing state $|f\rangle$", i.e.: $S_{fi} = \langle f|f\rangle_{\text{out}} = \langle f|S|i\rangle$.

In a theory without interaction, the S operator is the identity I. Generally, $S = I + iT$, where iT is the interaction term.

For Γ, we have $d\Gamma = \frac{1}{2m_R}\left(\prod_f \frac{d^3 p_f}{(2\pi)^3 (2E_f)}\right)(2\pi)^3 (2E_n) \cdot (2\pi)^4 \delta^4\left(p_R - \sum p_i\right) \cdot |\mathcal{M}_{R\to f}|^2$, where

- m_R: rest mass of the parent particle R (in the center-of-mass frame, the parent particle momentum is 0, energy $E_R = m_R c^2$, in natural units $E_R = m_R$)
- p_i, E_i: momentum and energy of the i-th decay product (satisfying the relativistic relation $E_i = \sqrt{m_i^2 + p_i^2}$)
- $\delta^4\left(P_R - \sum p_i\right)$: four-dimensional delta function, enforcing **energy-momentum conservation** in the decay process (non-zero probability only when total energy-momentum is conserved)
- Decay amplitude $\mathcal{M}_{R\to f}$

Additionally, we have the well-known Breit-Wigner formula

$$\mathcal{M}(E) \approx \frac{\sqrt{\Gamma_{\text{in}} \cdot \Gamma_{\text{out}}}}{E - E_R + i\Gamma/2}$$

describing the relation between Γ and $\mathcal{M}$.

1.8 Feynman Rules in QED

In Quantum Electrodynamics (QED), Feynman diagrams gain physical meaning. Particles are represented by lines: fermions by solid lines, photons by wavy lines, bosons by dashed lines, gluons by looped lines. The connection point of one line with another is called a vertex. The horizontal (or vertical) axis of a Feynman diagram usually represents the time axis, with positive direction to the right; the left side represents the initial state, the right side the final state. An arrow pointing rightward indicates positive lepton or baryon number for a fermion; an arrow pointing leftward indicates negative lepton or baryon number.

1.8.1 Spin

A particle's spin is its intrinsic angular momentum (independent of orbital angular momentum), described by the spin operator $\hat{\vec{S}}$ (for a particle with spin quantum number s, the spin magnitude is $|\hat{\vec{S}}| = \sqrt{s(s+1)}\hbar$). Helicity is defined as the projection of the particle's spin angular momentum onto its momentum direction:

$$\lambda = \frac{\hat{\vec{S}} \cdot \hat{\vec{p}}}{\hbar}$$

Particle type	Spin quantum number s	Possible helicity h values	Note (mass-helicity relation)
Electron, proton, quark	$1/2$	$+1/2, -1/2$	Massive particle (helicity can change via Lorentz transformation)
Photon, gluon	1	$+1, -1$	Massless particle (helicity is Lorentz invariant)
Higgs boson	0	0 (only value)	Spin 0, no spin-momentum orientation, helicity always 0

1.8.2 Feynman Rules

Introduce three single-particle states with definite momentum and helicity:

$$\left|\mathbf{p}^{+}, \lambda\right\rangle = \sqrt{2E_{\mathbf{p}}}\, a^{\dagger}_{\mathbf{p},\lambda} \left|0\right\rangle$$
$$\left|\mathbf{p}^{-}, \lambda\right\rangle = \sqrt{2E_{\mathbf{p}}}\, b^{\dagger}_{\mathbf{p},\lambda} \left|0\right\rangle$$

Figure 1.9: Symbolic representation of single-particle states

Clearly, we can define the contraction of scalar fields:

$$\boxed{\phi(x)|\mathbf{p}\ \rangle = e^{-ip\cdot x}, \quad \langle\mathbf{p}|\phi(x) = e^{-ip\cdot x}}$$

Thus we have the Feynman diagram:

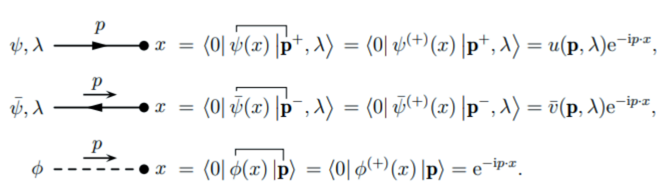
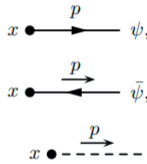


Figure 1.10: Feynman diagram for basic scattering process

Similarly,

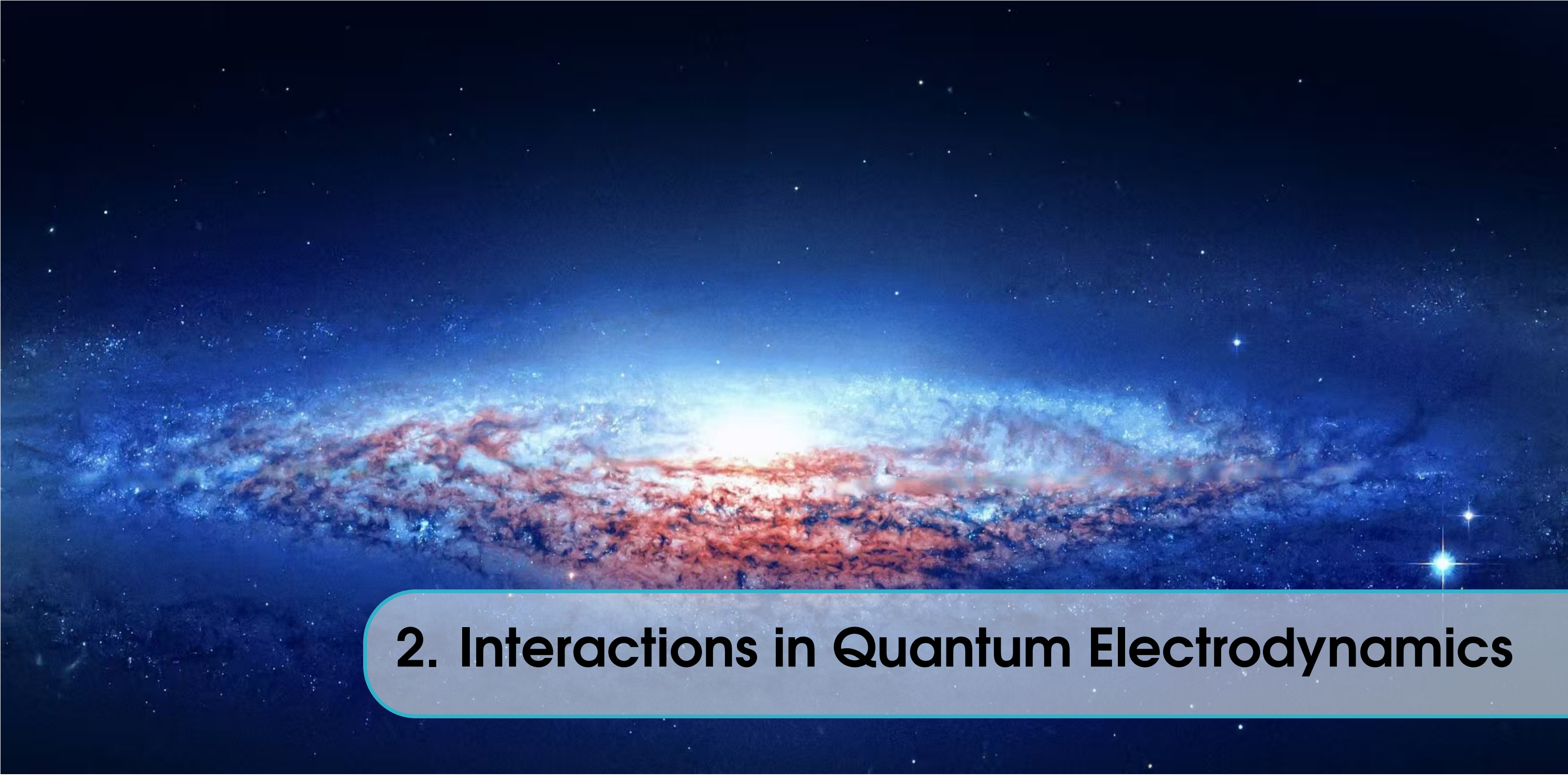


$$\psi, \lambda \;=\; \langle \mathbf{p}^+, \lambda |\, \bar{\psi}(x)\, |0\rangle = \langle \mathbf{p}^+, \lambda |\, \bar{\psi}^{(-)}(x)\, |0\rangle = \bar{u}(\mathbf{p}, \lambda) \mathrm{e}^{\mathrm{i}p\cdot x},$$

$$\bar{\psi}, \lambda \;=\; \langle \mathbf{p}^-, \lambda |\, \psi(x)\, |0\rangle = \langle \mathbf{p}^-, \lambda |\, \psi^{(-)}(x)\, |0\rangle = v(\mathbf{p}, \lambda) \mathrm{e}^{\mathrm{i}p\cdot x},$$

$$\phi \;=\; \langle \mathbf{p} |\, \phi(x)\, |0\rangle = \langle \mathbf{p} |\, \phi^{(-)}(x)\, |0\rangle = \mathrm{e}^{\mathrm{i}p\cdot x}.$$

Figure 1.11: Feynman diagram for other scattering processes



2. Interactions in Quantum Electrodynamics

2.1 Lagrangian Density of QED

Based on the discussion in the introduction, we can derive the Lagrangian density of QED:

$$\mathcal{L}_{QED} = \mathcal{L}_0 + \mathcal{L}_{int}$$

where $\mathcal{L}_0$ is the free particle term, which obviously includes the free fermion term and the free electromagnetic field term:

$$\mathcal{L}_0 = i\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi - \frac{1}{16\pi}F_{\mu\nu}F^{\mu\nu}$$

Next, we derive the interaction term $\mathcal{L}_{\text{int}}$. To make the QED Lagrangian density satisfy $U(1)$ gauge symmetry, i.e., remain invariant under the transformations:

$$\begin{aligned}
&\psi(x) \to \psi'(x) = e^{ie\theta(x)}\psi(x) \\
&\bar{\psi}(x) \to \bar{\psi}'(x) = \bar{\psi}(x)e^{-ie\theta(x)} \\
&A_\mu(x) \to A'_\mu(x) = A_\mu(x) - \partial_\mu\theta(x)
\end{aligned}$$

When taking the derivative of $\partial_\mu\psi'(x)$, because $\theta(x)$ depends on x, an additional phase gradient term appears:

$$\partial_\mu\psi'(x) = \partial_\mu\left[e^{ie\theta(x)}\psi(x)\right] = e^{ie\theta(x)}\left(ie\partial_\mu\theta(x)\cdot\psi(x) + \partial_\mu\psi(x)\right)$$

Substituting into the derivative term of the Dirac field:

$$i\bar{\psi}'\gamma^\mu\partial_\mu\psi' = i\bar{\psi}e^{-ie\theta}\gamma^\mu\cdot e^{ie\theta}\left(ie\partial_\mu\theta\cdot\psi + \partial_\mu\psi\right) = i\bar{\psi}\gamma^\mu\partial_\mu\psi - e\bar{\psi}\gamma^\mu\psi\cdot\partial_\mu\theta$$

We see an extra term $-e\bar{\psi}\gamma^\mu\psi\cdot\partial_\mu\theta$, causing $\mathcal{L}_\psi$ to no longer be invariant. This means the free fermion field alone cannot satisfy $U(1)$ symmetry. From $A_\mu(x) \to A'_\mu(x) = A_\mu(x) - \partial_\mu\theta(x)$, we achieve invariance of the total Lagrangian density $\mathcal{L}_{\text{int}} \to \mathcal{L}_{\text{int}} + e\bar{\psi}\gamma^\mu\psi\cdot\partial_\mu\theta$. Therefore the interaction term is

$$\mathcal{L}_{\text{int}} = -e\bar{\psi}\gamma^\mu\psi A_\mu$$

The pure electromagnetic field term without fermions already satisfies $U(1)$ gauge symmetry. In summary,

$$\mathcal{L}_{\text{QED}} = i\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi - \tfrac{1}{16\pi}F_{\mu\nu}F^{\mu\nu} - e\bar{\psi}\gamma^\mu\psi A_\mu$$

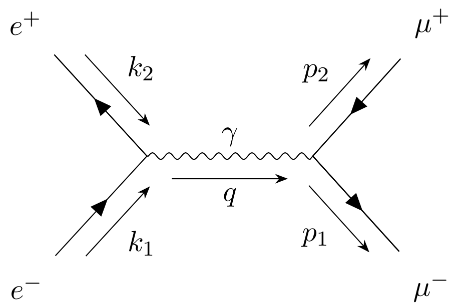
But this form is still not very elegant. We can introduce the covariant derivative $D_\mu = \partial_\mu - ieA_\mu(x)$, which clearly satisfies the transformation covariance $D'_\mu\psi'(x) = e^{ie\theta(x)}D_\mu\psi(x)$. Finally, we obtain

$$\boxed{\mathcal{L}_{\text{QED}} = i\bar{\psi}\gamma^\mu D_\mu\psi - m\bar{\psi}\psi - \frac{1}{16\pi}F_{\mu\nu}F^{\mu\nu}}$$

2.2 $e^+e^- \to \mu^+\mu^-$ Process

2.2.1 Amplitude and Cross Section Calculation for $e^+e^- \to \mu^+\mu^-$

Let's look at the Feynman diagram in Quantum Electrodynamics. According to the rules from the previous chapter, we clearly have:



We know $Q_e = Q_\mu = -1$, replace e^-, e^+, μ^-, μ^+ with $\lambda_1, \lambda_2, \lambda'_1, \lambda'_2$ respectively. For $e^+e^- \to \mu^+\mu^-$, k_1, k_2, p_1, p_2 are momenta, obtain amplitude:

$$\mathrm{i}\mathcal{M} = -\frac{\mathrm{i}e^2}{q^2}\overline{v}(k_2,\lambda_2)(\mathrm{i}e\gamma^\mu)u(k_1,\lambda_1)\frac{-\mathrm{i}g_{\mu\nu}}{q^2}\overline{u}(p_1,\lambda'_1)(\mathrm{i}e\gamma^\nu)v(p_2,\lambda'_2)$$

$$= -\frac{\mathrm{i}e^2}{q^2}\overline{v}(k_2,\lambda_2)\gamma^\mu u(k_1,\lambda_1)\overline{u}(p_1,\lambda'_1)\gamma_\mu v(p_2,\lambda'_2)$$

Photon is massless ($m_\gamma = 0$), no integral pole, so $= 0$, amplitude conjugate:

$$(\mathrm{i}\mathcal{M})^* = -\frac{\mathrm{i}e^2}{q^2}\overline{u}(k_1,\lambda_1)\gamma^\mu v(k_2,\lambda_2)\overline{v}(p_2,\lambda'_2)\gamma_\mu u(p_1,\lambda'_1)$$

From momentum conservation $q^2 = k_1^2 + k_2^2 + 2k_1 \cdot k_2 = t^\mu + p^\mu$, center-of-mass energy squared $E_{CM}^2 = q^2 = (p_1 + p_2)^2$, obtain amplitude modulus:

$$|\mathcal{M}|^2 = \frac{e^4}{E_{CM}^4}\left|\overline{v}(k_2,\lambda_2)\gamma^\mu u(k_1,\lambda_1)\overline{u}(p_1,\lambda'_1)\gamma_\mu v(p_2,\lambda'_2)\right|^2$$

$$= \frac{e^4}{E_{CM}^4}\mathrm{Tr}\left[\gamma^\mu(k_2 - m_e)\gamma^\nu(k_1 + m_e)\right]\mathrm{Tr}\left[\gamma_\mu(p_1 + m_\mu)\gamma_\nu(p_2 - m_\mu)\right]$$

Step 2 uses trace trick, the expressions inside the trace represent positive/negative electron/positron spinors respectively.

2.3 Trace Calculation for Dirac Matrices

Use trace techniques to simplify the above formula (for Dirac γ matrices):

$$\mathrm{Tr}(\gamma^\mu\gamma^\nu) = 4g^{\mu\nu}, \quad \mathrm{Tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = 4(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\rho}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\rho})$$

Easy to obtain:

$$\mathrm{Tr}[\gamma^\mu(\gamma^\nu)^*] = \mathrm{Tr}\left[\gamma^\mu(\gamma^\nu)^T\gamma^5\right]$$

$$= -\mathrm{Tr}\left[\gamma^5\gamma^\mu(\gamma^\nu)^T\right]$$

$$= -\mathrm{Tr}\left[(\gamma^\mu)^T(\gamma^\nu)^*\right]$$

Thus $\mathrm{Tr}[\gamma^\mu(\gamma^\nu)^*] = 0$. Also from $\gamma^\mu\gamma^\nu = 2g^{\mu\nu} - \gamma^\nu\gamma^\mu$, simplify:

$$\mathrm{Tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = 2g^{\mu\nu}\mathrm{Tr}(\gamma^\rho\gamma^\sigma) - 2g^{\mu\rho}\mathrm{Tr}(\gamma^\nu\gamma^\sigma) + 2g^{\mu\sigma}\mathrm{Tr}(\gamma^\nu\gamma^\rho) - \mathrm{Tr}(\gamma^\nu\gamma^\mu\gamma^\rho\gamma^\sigma)$$

$$= \mathrm{Tr}\left(2g^{\mu\nu}\gamma^\rho\gamma^\sigma - 2g^{\mu\rho}\gamma^\nu\gamma^\sigma + 2g^{\mu\sigma}\gamma^\nu\gamma^\rho - \gamma^\nu\gamma^\mu\gamma^\rho\gamma^\sigma\right)$$

From $\mathrm{Tr}(\gamma^\mu\gamma^\nu) = 4g^{\mu\nu}$:

$$2\mathrm{Tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = 8(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\rho}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\rho})$$

$$\mathrm{Tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = 4(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\rho}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\rho})$$

Thus:

$$\mathrm{Tr}\left[(k_2 - m_e)\gamma^\mu(k_1 + m_e)\gamma^\nu\right] = \mathrm{Tr}\left[k_{2\rho}\gamma^\rho\gamma^\mu k_{1\sigma}\gamma^\sigma\gamma^\nu - m_e^2\gamma^\mu\gamma^\nu\right]$$

$$= \mathrm{Tr}\left(k_{2\rho}k_{1\sigma}\gamma^\rho\gamma^\mu\gamma^\sigma\gamma^\nu\right) - m_e^2\mathrm{Tr}(\gamma^\mu\gamma^\nu)$$

From:

$$\mathrm{Tr}\left[k_{2\rho}k_{1\sigma}\gamma^\rho\gamma^\mu\gamma^\sigma\gamma^\nu\right] = k_{2\rho}k_{1\sigma}4(g^{\rho\mu}g^{\sigma\nu} - g^{\rho\sigma}g^{\mu\nu} + g^{\rho\nu}g^{\mu\sigma})$$

$$= 4(k_2^\mu k_1^\nu + k_2^\nu k_1^\mu - g^{\mu\nu}k_1 \cdot k_2)$$

First term becomes $4\left[k_2^\mu k_1^\nu + k_2^\nu k_1^\mu - g^{\mu\nu}(k_1 \cdot k_2 + m_e^2)\right]$, second term is $4\left[k_1^\mu k_2^\nu + k_2^\mu k_1^\nu - g^{\mu\nu}(p_1 \cdot p_2 + m_\mu^2)\right]$, so:

2.4 Momentum Conservation and Center-of-Mass Frame

Obtain $E_{CM} = 2(m_\mu^2 + p \cdot p_2)$, i.e.:

$$k_1 \cdot k_2 = \frac{E_{CM}^2}{2} - m_e^2, \quad p_1 \cdot p_2 = \frac{E_{CM}^2}{2} - m_\mu^2$$

From $k_1 + k_2 = p_1 + p_2$, we have $k_1 - p_1 = p_2 - k_2$, then:

$$(k_1 - p_1)^2 = m_e^2 + m_\mu^2 - 2k_1 \cdot p_1, \quad (p_2 - k_2)^2 = m_e^2 + m_\mu^2 - 2k_2 \cdot p_2$$

Finally:

$$k_2 \cdot p_2 = k_1 \cdot p_1 = k_1^\mu p_1^\mu - |k_1||p_1|\cos\theta = \frac{E_{CM}^2}{4}(1 - \beta_e\beta_\mu\cos\theta)$$

$$k_2 \cdot p_1 = k_1 \cdot p_2 = k_1^\mu p_2^\mu - |k_1||p_2| \cos\theta = \frac{E_{CM}^2}{4}(1 + \beta_e \beta_\mu \cos\theta)$$

Using momentum conservation, define fine-structure constant:

$$\alpha = \frac{e^2}{4\pi} \approx \frac{1}{137}$$

Obtain:

$$|\mathcal{M}|^2 = \frac{8e^4}{E_{CM}^2}\left[E_{CM}^2(\beta_e\beta_\mu\cos\theta)^2 + \beta_e^2(\beta_\mu\cos\theta) + m_e^2\left(\frac{E_{CM}^2}{2} - m_e^2\right) + m_\mu^2\left(\frac{E_{CM}^2}{2} - m_\mu^2\right) + 2m_e^2 m_\mu^2\right]$$

$$= \frac{8e^4}{E_{CM}^2}\left[E_{CM}^2(1 + \beta_e^2\cos^2\theta) + 4(m_e^2 + m_\mu^2)\right]$$

$$= 16\pi^2\alpha^2\left[1 + \beta_e^2\cos^2\theta + \frac{4(m_e^2 + m_\mu^2)}{E_{CM}^2}\right]$$

Since $|\mathcal{M}|^2 \propto \alpha^2$, this is α^2-order. Generally, in QED, Feynman diagram with n vertices is α^n-order (one vertex contributes one e factor).

2.5 Differential Scattering Cross Section

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{\alpha^2}{4E_{CM}^2}\left[1 + \beta_e^2\cos^2\theta + \frac{4(m_e^2 + m_\mu^2)}{E_{CM}^2}\right]$$

From $\int_0^\pi \mathrm{d}\theta \sin\theta = \int_{-1}^1 \mathrm{d}(\cos\theta) = 2$, $\int_0^\pi \mathrm{d}\theta \sin\theta\cos\theta = \int_{-1}^1 \cos\theta \mathrm{d}(\cos\theta) = \frac{1}{2}$, unpolarized cross section:

$$\sigma = \int_0^{2\pi} \mathrm{d}\phi \int_0^\pi \mathrm{d}\theta \sin\theta \frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{\pi\alpha^2}{E_{CM}^2}\left[2 + 2\beta_e^2 + \frac{8(m_e^2 + m_\mu^2)}{E_{CM}^2}\right]$$

$$= \frac{\pi\alpha^2}{E_{CM}^2}\left[3\left(1 - \frac{4m_e^2}{E_{CM}^2}\right) + \frac{12(m_e^2 + m_\mu^2)}{E_{CM}^2}\right]$$

$$= \frac{4\pi\alpha^2}{3E_{CM}^2}\left[1 + \frac{2(m_e^2 + m_\mu^2)}{E_{CM}^2} + \frac{E_{CM}^2}{4}\right]$$

$$= \frac{4\pi\alpha^2}{3E_{CM}^2}\left(1 + \frac{2m_e^2}{E_{CM}^2}\right)\left(1 + \frac{2m_\mu^2}{E_{CM}^2}\right)$$

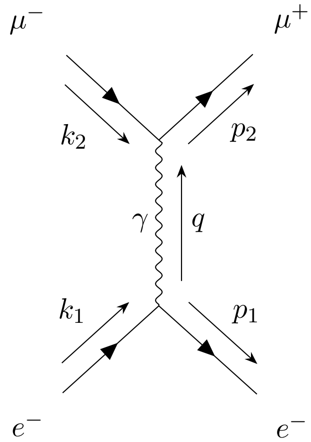
If $E_{CM} \gg m_e, m_\mu$:

$$\sigma(e^+e^- \to \mu^+\mu^-) = \frac{4\pi\alpha^2}{3E_{CM}^2}$$

2.5.1 Coulomb Scattering

Coulomb process describes virtual photon exchange between charged particles.

For e^-e^+ scattering ($e^-e^+ \to e^-e^+$), the Feynman diagram is:



Amplitude:

$$\mathrm{i}\mathcal{M} = -\mathrm{i}e^2(\mathrm{i}e)\gamma^\mu u(k)\frac{-\mathrm{i}g_{\mu\nu}}{q^2}(\mathrm{i}e)\gamma^\nu v(k')(\mathrm{i}e)\gamma^\rho u(p)(\mathrm{i}e)\gamma_\rho v(p')$$

From previous discussion:

$$|\mathcal{M}|^2 = \frac{8e^4}{q^4}\mathrm{Tr}\left[(p' + m_e)\gamma^\mu(k' + m_e)\gamma^\nu\right]\mathrm{Tr}\left[(p' + m_\mu)\gamma_\mu(k' + m_\mu)\gamma_\nu\right]$$

$$= \frac{8e^4}{q^4}\left[(k' \cdot p')(k' \cdot p) + (k' \cdot k)(p' \cdot p) - m_e^2(k' \cdot k) - m_\mu^2(k' \cdot p') + 2m_e^2 m_\mu^2\right]$$

Center-of-mass energy:

$$E_{CM} = E_e + E_\mu = E_e + \sqrt{m_\mu^2 + p_\mu^2} = E_e + \sqrt{E_e^2 - m_e^2 + m_\mu^2}$$

Step 1 uses $|p| = |p'|$, so:

$$E_e^2 - m_e^2 + m_\mu^2 = (E_{CM} - E_e)^2 = E_{CM}^2 - 2E_{CM}E_e + E_e^2$$

$$2E_{CM}E_e = E_{CM}^2 + m_e^2 - m_\mu^2$$

$$E_e = \frac{1}{2E_{CM}}(E_{CM}^2 + m_e^2 - m_\mu^2), \quad E_\mu = \frac{1}{2E_{CM}}(E_{CM}^2 + m_\mu^2 - m_e^2)$$

Define Källen function:

$$\lambda(x, y, z) = (x - y - z)^2 - 4yz = x^2 + y^2 + z^2 - 2xy - 2xz - 2yz$$

Obtain:

$$|p'|^2 = E_e^2 - m_e^2 = \frac{1}{4E_{CM}^2}(E_{CM}^2 + m_e^2 - m_\mu^2)^2 - m_e^2$$

$$= \frac{1}{4E_{CM}^2}\left[E_{CM}^4 + m_e^4 + m_\mu^4 + 2E_{CM}^2 m_e^2 - 2E_{CM}^2 m_\mu^2 - 2m_e^2 m_\mu^2 - 4E_{CM}^2 m_e^2\right]$$
$$= \frac{1}{4E_{CM}^2}\left[E_{CM}^4 + m_e^4 + m_\mu^4 - 2E_{CM}^2 m_e^2 - 2E_{CM}^2 m_\mu^2 - 2m_e^2 m_\mu^2\right]$$
$$= \frac{1}{4E_{CM}^2}\lambda(E_{CM}^2, m_e^2, m_\mu^2)$$

Where $Q = |p| = |p'| = |k| = |k'| = \frac{1}{2E_{CM}}\lambda(E_{CM}^2, m_e^2, m_\mu^2) = \frac{\sqrt{\lambda}}{2E_{CM}}\left(1, \frac{m_e^2}{E_{CM}^2}, \frac{m_\mu^2}{E_{CM}^2}\right)$
With Q as momentum transfer, $k_1' = p_1^0 = E_e$, $k_2' = p_2^0 = E_\mu$:

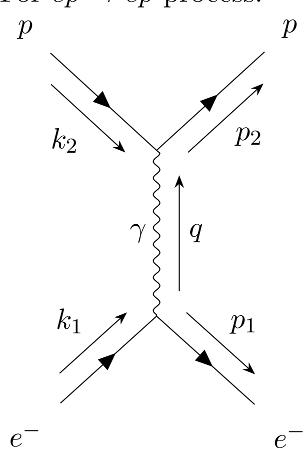
$$k_1' \cdot p_1' = k_1' p_1' - |k_1'||p_1'|\cos\theta = E_e^2 - Q^2\cos\theta, \quad k_2' \cdot p_2' = E_\mu^2 - Q^2\cos\theta$$

$$k_1' \cdot p_2' = E_e E_\mu + Q^2\cos\theta = k_2' \cdot p_1', \quad k_1' \cdot k_2' = E_e E_\mu + Q^2$$
$$q^2 = (k_1 - p_1')^2 = 2m_e^2 - 2k_1 \cdot p_1' = 2(m_e^2 - E_e^2 + Q^2\cos\theta) = -2Q^2(1-\cos\theta)$$

Thus:

$$|\mathcal{M}|^2 = \frac{8e^4}{q^4}\left[(k\cdot P)(k\cdot k) + (k\cdot k)(P\cdot P) - m_e^2(k\cdot k) - m_p^2(k\cdot P) + 2m_e^2 m_p^2\right]$$

For $ep \to ep$ process:



Electron mass $\ll$ proton mass, electron initial/final velocity $v = \frac{Q}{E}$ (total momentum Q, electron energy E), proton momentum before/after is 0 (at rest), i.e.:

$$k^\mu = (E, k), \quad k'^\mu = (m_p, 0), \quad p^\mu = (E, P), \quad p'^\mu = (m_p, 0)$$

Since $E_p = 0 + m_p^2$, and:

$$k \cdot P = E^2 Q^2 \cos\theta = E^2(1 - v^2\cos\theta), \quad k \cdot P = m_p^2$$

$$q^2 = (k - P)^2 = 2m_e^2 - 2E^2 + 2Q^2\cos\theta = -2Q^2(1-\cos\theta)$$

Using previous formula, obtain:

$$|\mathcal{M}|^2 = \frac{8e^4}{q^4}\left[(k\cdot P)(k\cdot k) + (k\cdot k)(P\cdot P) - m_e^2(k\cdot k) - m_p^2(k\cdot P) + 2m_e^2 m_p^2\right]$$

After substitution:

$$|\mathcal{M}|^2 = \frac{8e^4}{4Q^4(1-\cos\theta)^2}\left[m_p^2E^2 + m_p^2E^2 - m_e^2m_p^2 - m_p^2E^2(1-v^2\cos\theta) + 2m_e^2m_p^2\right]$$

$$= \frac{32\pi^2\alpha^2}{Q^4(1-\cos\theta)^2}\left(m_p^2E^2 + m_p^2E^2v^2\cos\theta + m_e^2m_p^2\right)$$

$$= \frac{32\pi^2\alpha^2}{Q^4(1-\cos\theta)^2}\left[m_p^2E^2 + m_p^2E^2v^2\cos\theta + m_p^2E^2(1-v^2)\right]$$

$$= \frac{832\pi^2\alpha^2m_p^2}{v^2Q^4(1-\cos\theta)^2}\left[2 - v^2(1-\cos\theta)\right]$$

$$= \frac{16\pi^2\alpha^2m_p^2}{v^2Q^4\sin^4\frac{\theta}{2}}\left(1 - v^2\sin^2\frac{\theta}{2}\right)$$

2.5.2 Differential Scattering Cross Section (Mott/Rutherford)

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{|\mathcal{M}|^2}{64\pi^2m_p^2} = \frac{\alpha^2}{4v^2Q^4\sin^4\frac{\theta}{2}}\left(1 - v^2\sin^2\frac{\theta}{2}\right)$$

Since $E_eE_p|p-p'| = E_mp\frac{Q}{E} = Qm_p$, and $Q \ll m_p$, this formula is called Mott formula; in classical case, it reduces to Rutherford formula:

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{\alpha^2}{4m_e^2v^4\sin^4\frac{\theta}{2}}$$

2.5.3 Predicting Particle Collider Experiments

We naturally ask: we've calculated a bunch of things, but what is it actually used for? Let's discuss its practical significance. The size of the cross section σ corresponds to the number of reactions per unit time. Define the **reaction rate** $R = \sigma \cdot L$, where L is the luminosity of the Large Electron-Positron Collider (LEP), expressed in $\mathrm{cm}^{-2}\mathrm{s}^{-1}$. If N particle pairs collide per unit time, we can compute the reaction probability per particle pair: $P = \frac{R}{N}$. From $L = \frac{N}{S} \implies N = L \cdot S$, where S is the beam cross-sectional area, we get:

$$P = \frac{\sigma L}{LS} = \frac{\sigma}{S}$$

2.5.4 Exercise 3.1.1

Calculate the probability of $e^+e^- \to \mu^+\mu^-$ occurring when $E = 91$ GeV (the Z boson peak energy) and $S_1 = 10^{-10}$ m^2.

2.6 General Scattering and Interaction

The scattering matrix is defined as:

$$S = \lim_{t\to+\infty}\lim_{t_0\to-\infty} U(t,t_0) = \mathbb{I} + iT$$

where U is the time-evolution operator. The matrix element between initial state $|i\rangle$ and final state $|f\rangle$ is:

$$S_{fi} = \langle f|S|i\rangle = \langle f|i\rangle + iT_{fi}$$

Without interaction, $S = \mathbb{I}$ is the unit matrix, ensuring momentum conservation: $iT_{fi} = \langle f|iT|i\rangle = (2\pi)^4\delta^{(4)}(p_i - p_f)i\mathcal{M}$.

Consider a 2-body initial state (particles A and B) to an n-body final state:

$$|i\rangle = \sqrt{4E_AE_B}a^\dagger_{\mathbf{p}_A}a^\dagger_{\mathbf{p}_B}|0\rangle, \quad |f\rangle = \left(\prod_{j=1}^{n}\sqrt{2E_j}a^\dagger_{\mathbf{p}_j}\right)|0\rangle$$

Normalization: $\mathcal{N}\tilde{V} = (2\pi)^3\delta^{(3)}(0) = \int d^3x$, and $\tilde{T}\tilde{V} = \int d^4x$. Then,

$$\langle i|i\rangle = 4E_AE_B\tilde{V}^2, \quad \langle f|f\rangle = \prod_{j=1}^{n}(2E_j\tilde{V})$$

The transition probability is $P_{fi} = \frac{|\langle f|T|i\rangle|^2}{\langle i|i\rangle\langle f|f\rangle} = \frac{|T_{fi}|^2}{\langle i|i\rangle\langle f|f\rangle}$. Due to momentum conservation $p^\mu_A + p^\mu_B = \sum_{j=1}^{n} p^\mu_j$, integration over all final-state particles is required for the total transition probability.

2.6.1 Decay Width and Phase Space

The decay width Γ_f, lifetime τ, branching ratio, and partial width are discussed. The reaction cross-section involves the differential cross-section $d\sigma$ and phase space elements. The Møller velocity is defined as:

$$v_{\text{Møller}} = \frac{1}{E_AE_B}\sqrt{(\mathbf{p}_A\cdot\mathbf{p}_B)^2 - m_A^2m_B^2}$$

The cross-section is then $\sigma = \frac{R\tilde{V}}{v_{\text{Møller}}}$. The n-body invariant phase space is:

$$\int d\Pi_n = \left(\prod_{i=1}^{n}\int\frac{d^3p_i}{(2\pi)^3 2E_i}\right)(2\pi)^4\delta^{(4)}(p_A + p_B - \sum_{i=1}^{n}p_i)$$

If the final state contains n_k identical particles of type k, a symmetry factor $\mathcal{S} = \prod_k n_k!$ must be included, except in the differential cross-section $d\sigma$. Thus,

$$\sigma = \frac{1}{\mathcal{S}}\int d\sigma = \frac{1}{\mathcal{S}}\frac{1}{4E_AE_Bv_{\text{Møller}}}\int d\Pi_n|\mathcal{M}|^2$$

For two-body scattering,

$$\sigma = \frac{1}{\mathcal{S}}\frac{1}{4E_AE_B|v_A - v_B|}\frac{|\mathbf{p}_f|}{16\pi^2\sqrt{s}}\int d\Omega|\mathcal{M}|^2$$

For particle decay, the number of particles changes as $dN = -\Gamma N dt$, leading to $N(t) = N(0)e^{-\Gamma t}$, which follows an exponential distribution with probability density $P(t) = \Gamma e^{-\Gamma t}$. The lifetime is $\tau = 1/\Gamma$, and Γ is the decay width. The branching ratio for decay $i \to f$ is B_f, and the partial width is $\Gamma_f = \Gamma B_f$. Thus,

$$\Gamma_f = \frac{1}{\mathcal{S}}\frac{1}{2m_A}\int d\Pi_n$$

where there is no particle B, i.e., $p_B = 0$ in $\int d\Pi_n$.

2.7 Mandelstam Variables and Ward Identities

Mandelstam variables are defined as:

$$s = (k_1 + k_2)^2 = (p_1 + p_2)^2, \quad t = (k_1 - p_1)^2 = (k_2 - p_2)^2, \quad u = (k_1 - p_2)^2 = (k_2 - p_1)^2$$

These correspond to the Feynman diagrams known as the s-channel, t-channel, and u-channel.

The QED interaction Lagrangian is $L_{\text{int}} = -eA_\mu \bar{\psi}\gamma^\mu\psi = -A_\mu J^\mu$, with the Noether current $J^\mu = e\bar{\psi}\gamma^\mu\psi$ satisfying $\partial_\mu J^\mu_{\text{em}} = 0$.

Consider the process with an outgoing photon. The amplitude is $i\mathcal{M}(p) = i\mathcal{M}^\mu(p)\varepsilon^*_\mu(p)$. Using $\partial_\mu J^\mu = 0$ and integrating by parts, we derive the Ward identity: $p_\mu \mathcal{M}^\mu(p) = 0$.

The photon polarization sum is:

$$\sum_{\text{spins}} \varepsilon^*_\mu(p)\varepsilon_\nu(p) = -g_{\mu\nu} - \frac{p_\mu p_\nu}{(p \cdot n)} + \frac{p_\mu n_\nu + p_\nu n_\mu}{(p \cdot n)}, \quad \text{where } n^\mu = (1, 0, 0, 0).$$

Using the Ward identity, in QED, $\sum_{\text{spins}} \varepsilon^*_\mu(p)\varepsilon_\nu(p)$ simplifies to $-g_{\mu\nu}$.

2.8 Compton Scattering and Klein-Nishina Formula

Compton scattering refers to $\gamma e^- \to \gamma e^-$. At tree level, the amplitude is:

$$\begin{aligned}
i\mathcal{M} &= \varepsilon^*_\mu(p_2)\bar{u}(p_1)(ie\gamma^\mu)\frac{i(\not{k}_1 + \not{k}_2 + m_e)}{(k_1 + k_2)^2 - m_e^2}(ie\gamma^\nu)u(k_1)\varepsilon_\nu(k_2) \\
&\quad + \varepsilon_\nu(k_2)\bar{u}(p_1)(ie\gamma^\mu)\frac{i(\not{k}_1 - \not{p}_1 + m_e)}{(k_1 - p_1)^2 - m_e^2}(ie\gamma^\mu)u(k_1)\varepsilon^*_\mu(p_2) \\
&= -ie^2\varepsilon^*_\mu(p_2)\varepsilon_\nu(k_2)\bar{u}(p_1)\left[\frac{\gamma^\mu(\not{k}_1 + \not{k}_2 + m_e)\gamma^\nu}{(k_1 + k_2)^2 - m_e^2} + \frac{\gamma^\nu(\not{k}_1 - \not{p}_1 + m_e)\gamma^\mu}{(k_1 - p_1)^2 - m_e^2}\right]u(k_1)
\end{aligned}$$

On-shell conditions: $k_1^2 = m_e^2$, $k_2^2 = p_1^2 = 0$, so

$$(k_1 + k_2)^2 - m_e^2 = 2k_1 \cdot k_2, \quad (k_1 - p_1)^2 - m_e^2 = -2k_1 \cdot p_2$$

Using $\gamma^\mu \not{p} = p_\nu\gamma^\mu\gamma^\nu = p_\nu(2g^{\mu\nu} - \gamma^\nu\gamma^\mu) = 2p^\mu - \not{p}\gamma^\mu$, we have $\not{k}_1\gamma^\nu = 2k_1^\nu - \gamma^\nu \not{k}_1$. From the Dirac equation, $(\not{k}_1 - m_e)u(k_1) = 0$, so

$$(\not{k}_1 + m_e)\gamma^\nu u(k_1) = 2k_1^\nu u(k_1) - \gamma^\nu(\not{k}_1 - m_e)u(k_1) = 2k_1^\nu u(k_1)$$

Thus,

$$i\mathcal{M} = -ie^2\varepsilon^*_\mu(p_2)\varepsilon_\nu(k_2)\bar{u}(p_1)\left[\frac{\gamma^\mu \not{k}_1\gamma^\nu + 2\gamma^\mu k_1^\nu}{2k_1 \cdot k_2} + \frac{\gamma^\nu \not{k}_1\gamma^\mu - 2\gamma^\nu k_1^\mu}{2k_1 \cdot p_2}\right]u(k_1)$$

The complex conjugate is:

$$(i\mathcal{M})^* = ie^2\varepsilon_\mu(p_2)\varepsilon^*_\nu(k_2)\bar{u}(k_1)\left[\frac{\gamma^\nu \not{k}_1\gamma^\mu + 2\gamma^\mu k_1^\nu}{2k_1 \cdot k_2} + \frac{\gamma^\mu \not{k}_1\gamma^\nu - 2\gamma^\mu k_1^\mu}{2k_1 \cdot p_2}\right]u(p_1)$$

The squared amplitude, averaged over initial spins and summed over final spins, is:

$$\overline{|\mathcal{M}|^2} = \frac{1}{4}\sum_{\text{spins}} |\mathcal{M}|^2 = \frac{e^4}{16}\left[\frac{A}{(k_1\cdot k_2)^2} + \frac{B+C}{(k_1\cdot k_2)(k_1\cdot p_2)} + \frac{D}{(k_1\cdot p_2)^2}\right]$$

Using the Ward identity, $\sum_{\text{spins}} \varepsilon^*_\mu(p)\varepsilon_\nu(p)$ is replaced by $-g_{\mu\nu}$, and trace techniques simplify the expressions. The Mandelstam variables yield $B = C$, with:

$$\begin{aligned} A &= 32[(k_1\cdot k_2)(k_1\cdot p_2) + m_e^2 k_1\cdot k_2 + m_e^4] \\ B = C &= 16(-m_e^2 k_1\cdot k_2 + m_e^2 k_1\cdot p_2 - 2m_e^4) \\ D &= 32[(k_1\cdot k_2)(k_1\cdot p_2) - m_e^2 k_1\cdot p_2 + m_e^4] \end{aligned}$$

Therefore,

$$\overline{|\mathcal{M}|^2} = 2e^4\left[\frac{k_1\cdot p_2}{k_1\cdot k_2} + \frac{k_1\cdot k_2}{k_1\cdot p_2} + 2m_e^2\left(\frac{1}{k_1\cdot k_2} - \frac{1}{k_1\cdot p_2}\right) + m_e^4\left(\frac{1}{k_1\cdot k_2} - \frac{1}{k_1\cdot p_2}\right)^2\right]$$

Using the angular frequency $\omega = E = 2\pi/\lambda = |\vec{k}_2|$, and from momentum conservation:

$$m_e^2 = p_1^2 = (k_1 + k_2 - p_2)^2 = m_e^2 + 2m_e(\omega - \omega') - 2\omega\omega'(1-\cos\theta)$$

$$(p_1^0)^2 = m_e^2 + |\vec{p}_1|^2 = m_e^2 + |\vec{k}_2 - \vec{p}_2|^2 = m_e^2 + \omega^2 + \omega'^2 - 2\omega\omega'\cos\theta$$

Thus, $\omega - \omega' = \frac{\omega\omega'}{m_e}(1-\cos\theta)$, and $\frac{1}{\omega'} - \frac{1}{\omega} = \frac{\omega-\omega'}{\omega'\omega} = \frac{1-\cos\theta}{m_e}$.

The Compton wavelength shift equation is presented as:

$$\Delta\lambda = \lambda' - \lambda = 2\pi\left(\frac{1}{\omega'} - \frac{1}{\omega}\right) = \frac{2\pi}{m_e}(1-\cos\theta)$$

The derivation employs center-of-momentum system momentum components:

$$k_1^\mu = (m_e, 0), \quad k_2^\mu = (\omega, \vec{k}_2), \quad p_1^\mu = (p_1^0, \vec{p}_1), \quad p_2^\mu = (\omega', \vec{p}_2)$$

Key momentum products are calculated:

$$k_1\cdot k_2 = m_e\omega, \quad k_1\cdot p_2 = m_e\omega'$$

The differential cross-section is derived as:

$$\begin{aligned} \left|\frac{d\sigma}{d\cos\theta}\right. &= \frac{\omega'^2}{32\pi m_e^2\omega^2}\frac{1}{|\mathcal{M}|^2} \\ &= \frac{2e^4\omega'^2}{32\pi m_e^2\omega^2}\left(\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - \sin^2\theta\right) \quad \text{which is the Klein-Nishina formula} \\ \frac{d\sigma}{d\cos\theta} &= \frac{\pi\alpha^2\omega'^2}{m_e^2\omega^2}\left(\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - \sin^2\theta\right) \\ &= 2e^4\left(\frac{\omega'}{\omega} + \frac{\omega}{\omega'} + \cos^2\theta - 1\right) = 2e^4\left(\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - \sin^2\theta\right) \end{aligned}$$

The analysis extends to two-body phase space integration:

$$\begin{aligned}\int d\Pi_2 &= \int \frac{d^3p_1}{(2\pi)^3 2E_1}\frac{d^3p_2}{(2\pi)^3 2E_2}(2\pi)^4\delta^{(4)}(k_1+k_2-p_1-p_2)\\ &= \int \frac{d^3p_2}{(2\pi)^2 4E_2 p_1^0}\delta(m_e+\omega-p_1^0-\omega')\end{aligned}$$

The integration proceeds by first integrating over p_1 and substituting p_1^0:

$$\begin{aligned}\int d\Pi_2 &= \int \frac{d\omega'\omega'^2 d\cos\theta}{16\pi^2\omega' p_1^0}\delta\left(m_e+\omega-\sqrt{m_e^2+\omega^2+\omega'^2-2\omega\omega'\cos\theta}-\omega'\right)\\ &= \int \frac{d\cos\theta\omega'}{8\pi p_1^0}\left|\frac{\partial}{\partial\omega'}\left(\sqrt{m_e^2+\omega^2+\omega'^2-2\omega\omega'\cos\theta}+\omega'\right)\right|^{-1}\\ &= \int \frac{d\cos\theta\omega'}{8\pi p_1^0}\left|\frac{2\omega'-2\omega\cos\theta}{2p_1^0}+1\right|^{-1} = \int\frac{d\cos\theta}{8\pi}\frac{\omega'}{\omega'+p_1^0-\omega\cos\theta}\\ &= \int \frac{d\cos\theta}{8\pi}\frac{\omega'}{m_e+\omega(1-\cos\theta)} = \int d\cos\theta\frac{\omega'^2}{8\pi m_e\omega}\end{aligned}$$

The first step uses $d^3p_2 = \omega'^2 d\omega' d\Omega$, and the solid angle is $d\Omega = \sin\theta d\theta d\phi = d\phi d\cos\theta$. The second step uses $\delta[f(x)] = \sum_i \delta(x-x_i)/|f'(x_i)|$, and energy conservation $p_1^0+\omega' = m_e+\omega$ is also used.

The cross-section is then:

$$\sigma = \frac{1}{4E_AE_B|v_A-v_B|}\int d\Pi_2|\overline{\mathcal{M}}|^2 = \frac{1}{4k_1^0k_2^0|k_1^0-k_2^0|}\int d\cos\theta\frac{\omega'^2}{8\pi m_e\omega}|\overline{\mathcal{M}}|^2 = \int d\cos\theta\frac{\omega'^2}{8\pi m_e^2\omega^2}|\overline{\mathcal{M}}|^2$$

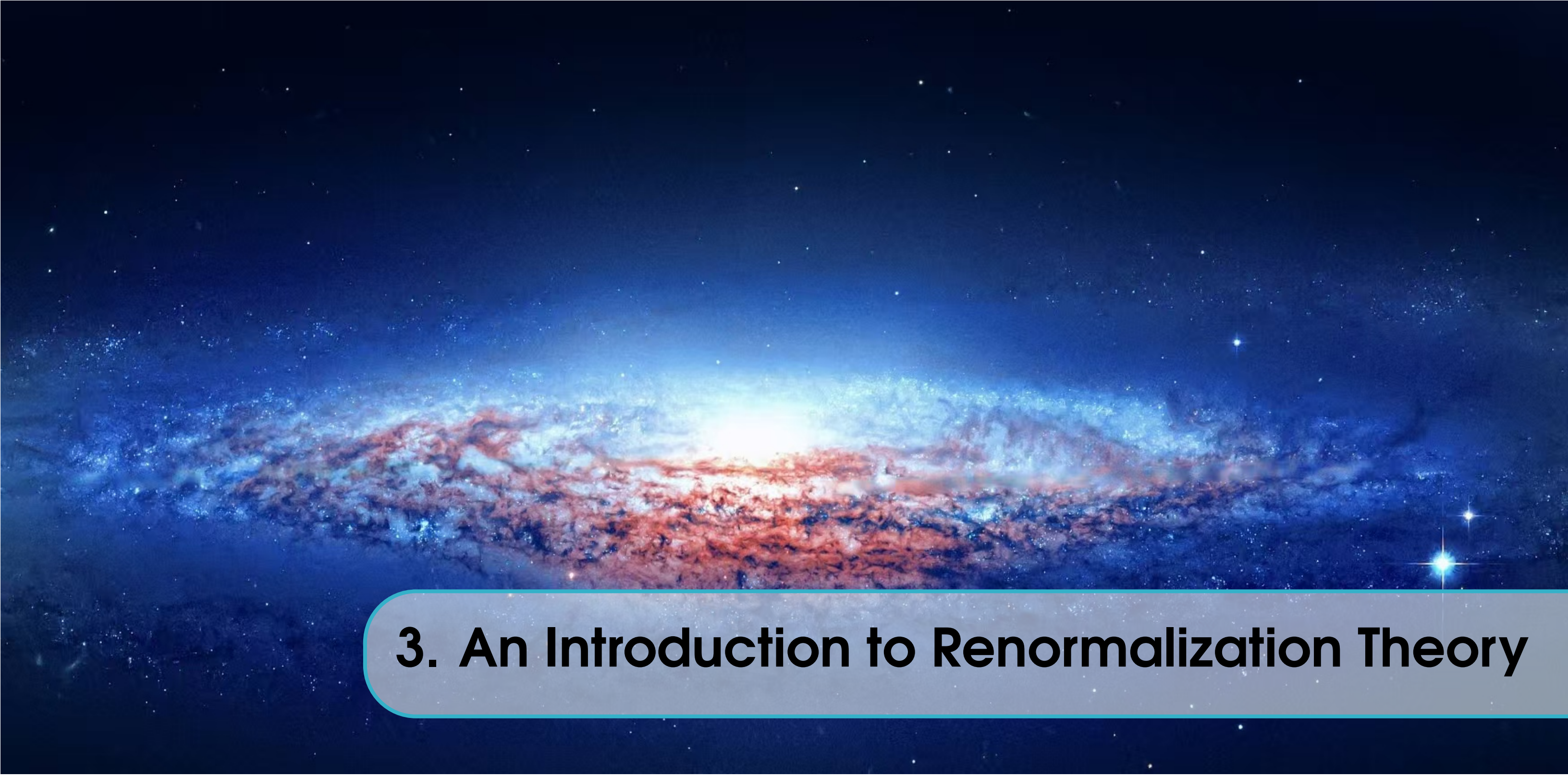
Therefore,

$$\frac{d\sigma}{d\cos\theta} = \frac{\omega'^2}{32\pi m_e^2\omega^2}|\mathcal{M}|^2 = \frac{2e^4\omega'^2}{32\pi m_e^2\omega^2}\left(\frac{\omega'}{\omega}+\frac{\omega}{\omega'}-\sin^2\theta\right)$$

which is the Klein-Nishina formula. In the non-relativistic limit, $\omega'/\omega = 1$, yielding the Thomson formula:

$$\frac{d\sigma}{d\cos\theta} = \frac{\pi\alpha^2}{m_e^2}(1+\cos^2\theta)$$

$$\sigma = \int_{-1}^{1} d\cos\theta\frac{\pi\alpha^2}{m_e^2}(1+\cos^2\theta) = \frac{8\pi\alpha^2}{3m_e^2}$$



3. An Introduction to Renormalization Theory

4. Standard Model and Grand Unified Theory

4.1 Elementary Particles of Standard Model

As we all know,fermions make up matter, while bosons mediate the interactions between fermions. The universe is made up of extremely tiny fragments of matter. Physicists call these fragments particles. As far as we know, there are many different types of particles that cannot be broken down any further. They are the most fundamental objects in the universe. Physicists refer to these fundamental objects as elementary particles. You may already be familiar with some of them; for example, the electron is an elementary particle. In addition, there are forces between these elementary particles that can attract or repel each other. Under appropriate conditions, two particles can even annihilate each other, leaving only energy behind. These forces are known as "interactions". Similarly, you might have learned about several fundamental interactions in kindergarten, such as the electromagnetic force. Of course, this is not the only one. The four fundamental forces are gravity, electromagnetism, the strong nuclear force, and the weak nuclear force, all of which are mediated by bosons.

The Higgs boson is the boson we know less about; it endows both bosons and fermions with mass through spontaneous symmetry breaking. Gluons exchange color charges between quarks, leading to quark confinement in protons and neutronsboth composed of quarksand thus forming atomic nuclei. The W and Z bosons mediate the weak interaction between fermions. Since light is an electromagnetic wave, it naturally acts as the mediator of the electromagnetic force. Last but not least, there is the graviton; although it is not included in the Standard Model and has little effect at the microscopic scale, it remains a crucially important force at the macroscopic level.

Here are known particles of the standard model:

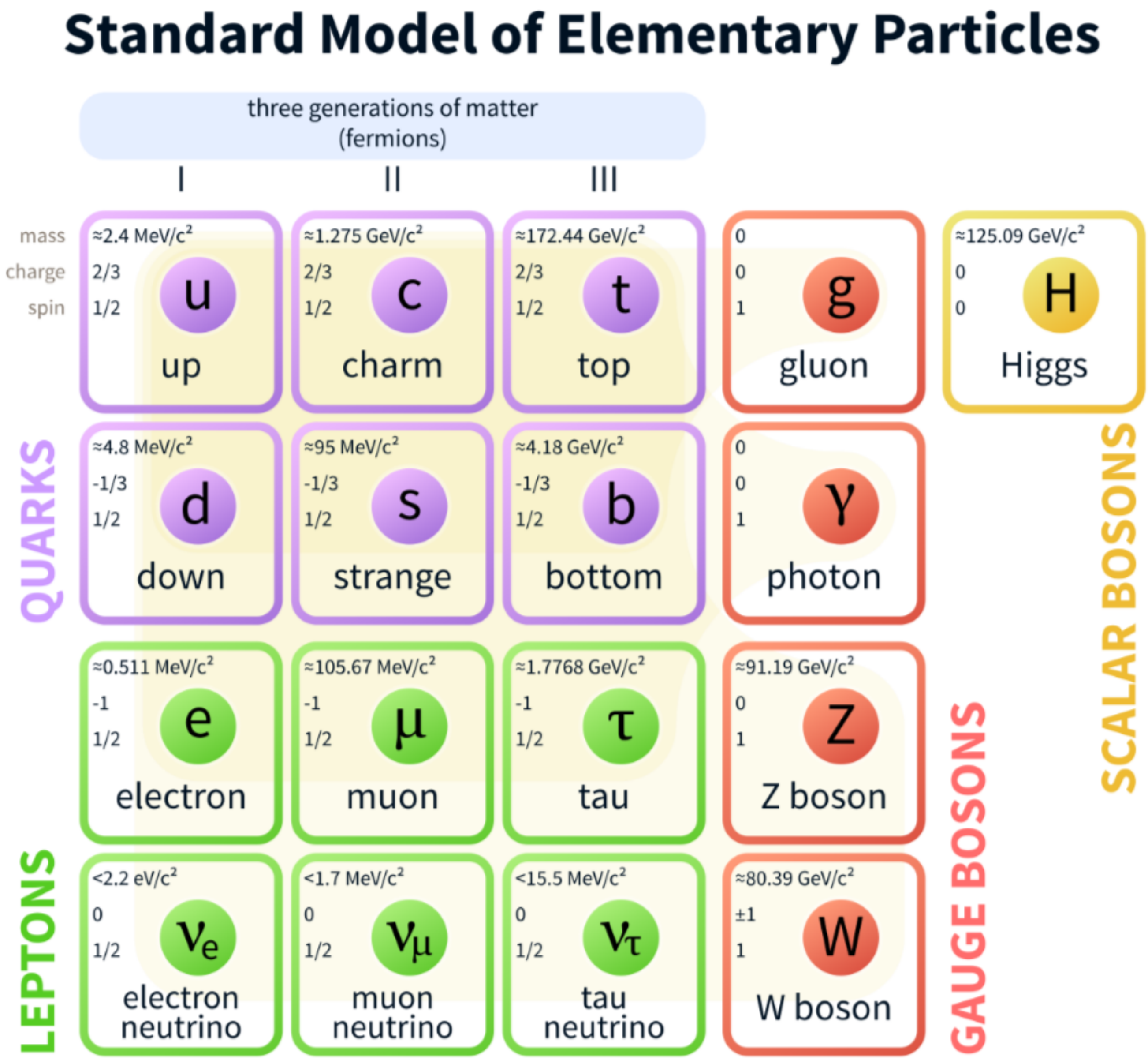


Figure 4.1: Standard Model

To be more specific, each elementary particle has unique properties that define its role in the universe. Fermions, the building blocks of matter, follow the Pauli Exclusion Principleno two identical fermions can occupy the same quantum stateensuring the stability of matter structures like atoms. They fall into two categories: quarks and leptons. Quarks carry "color charge" (the source of strong interaction) and exist only in bound states (such as protons and neutrons) due to quark confinement; there are six types (flavors) of quarks, each with different masses, from the lightest up and down quarks to the heaviest top quark. Leptons, including electrons, muons, tauons and their corresponding neutrinos, do not carry color charge and can exist freely; neutrinos are extremely light, almost massless, and only interact weakly with other particles, making them difficult to detect. Bosons, the mediators of interactions, do not follow the Pauli Exclusion Principle and can occupy the same quantum state. The Higgs boson, with a large mass, is responsible for the spontaneous symmetry breaking of the electroweak force, giving mass to other elementary particleswithout it, W and Z bosons would be massless, and the weak interaction would not have a short range. Gluons, massless and chargeless, come in eight types and couple strongly to quarks' color charge; their continuous exchange binds quarks tightly together, forming hadrons (protons, neutrons, etc.) and sustaining the stability of atomic nuclei. W bosons (W^+, W^-) carry electric charge and Z bosons are neutral; both have large masses, which limits the weak interaction to an extremely short range (about 10^{-18} meters), and they are key to processes like beta decay and neutrino scattering. Photons, massless and chargeless, mediate the electromagnetic force, governing phenomena such as atomic bonding, light propagation, and electric/magnetic interactions. The graviton, though unconfirmed experimentally, is hypothesized to be massless and mediate the gravitational force, governing macroscopic phenomena like planetary motion and cosmic expansion.

Back to the point,we have Standard Model Lagrangian density

$$\boxed{\mathcal{L}_{\text{SM}} = \mathcal{L}_{\text{gauge}} + \mathcal{L}_{\text{fermion}} + \mathcal{L}_{\text{Higgs}} + \mathcal{L}_{\text{Yukawa}}}$$

Where

$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4} G^a_{\mu\nu} G^{a\mu\nu} - \frac{1}{4} W^i_{\mu\nu} W^{i\mu\nu} - \frac{1}{4} B_{\mu\nu} B^{\mu\nu}$$

$$G^a_{\mu\nu} = \partial_\mu G^a_\nu - \partial_\nu G^a_\mu + g_s f^{abc} G^b_\mu G^c_\nu$$

$$W^i_{\mu\nu} = \partial_\mu W^i_\nu - \partial_\nu W^i_\mu + g \varepsilon^{ijk} W^j_\mu W^k_\nu$$

$$B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu$$

$$\mathcal{L}_{\text{fermion}} = \sum_\psi \bar{\psi}\, i\gamma^\mu D_\mu \psi = \bar{Q}_{L\alpha} i\gamma^\mu D_\mu Q_{L\alpha} + \bar{u}_{R\alpha} i\gamma^\mu D_\mu u_{R\alpha} + \bar{d}_{R\alpha} i\gamma^\mu D_\mu d_{R\alpha} + \bar{L}_{L\alpha} i\gamma^\mu D_\mu L_{L\alpha} + \bar{l}_{R\alpha} i\gamma^\mu D_\mu l_{R\alpha}$$

$$Q_L = \begin{pmatrix} u_L \\ d_L \end{pmatrix}, \quad L_L = \begin{pmatrix} \nu_L \\ l_L \end{pmatrix}$$

$$D_\mu = \partial_\mu - i g_s T^a G^a_\mu - i g T^i W^i_\mu - i g' \frac{Y}{2} B_\mu$$

$$\mathcal{L}_{\text{Higgs}} = |D_\mu \Phi|^2 - \mu^2 \Phi^\dagger \Phi - \lambda (\Phi^\dagger \Phi)^2$$

$$D_\mu \Phi = \left(\partial_\mu - i g T^i W^i_\mu - i g' \frac{Y}{2} B_\mu \right) \Phi, \quad Y = +1$$

$$\mathcal{L}_{\text{Yukawa}} = -\bar{Q}_{L\alpha} Y_u^{\alpha\beta} \tilde{\Phi} u_{R\beta} - \bar{Q}_{L\alpha} Y_d^{\alpha\beta} \Phi d_{R\beta} - \bar{L}_{L\alpha} Y_l^{\alpha\beta} \Phi l_{R\beta} + \text{h.c.}$$

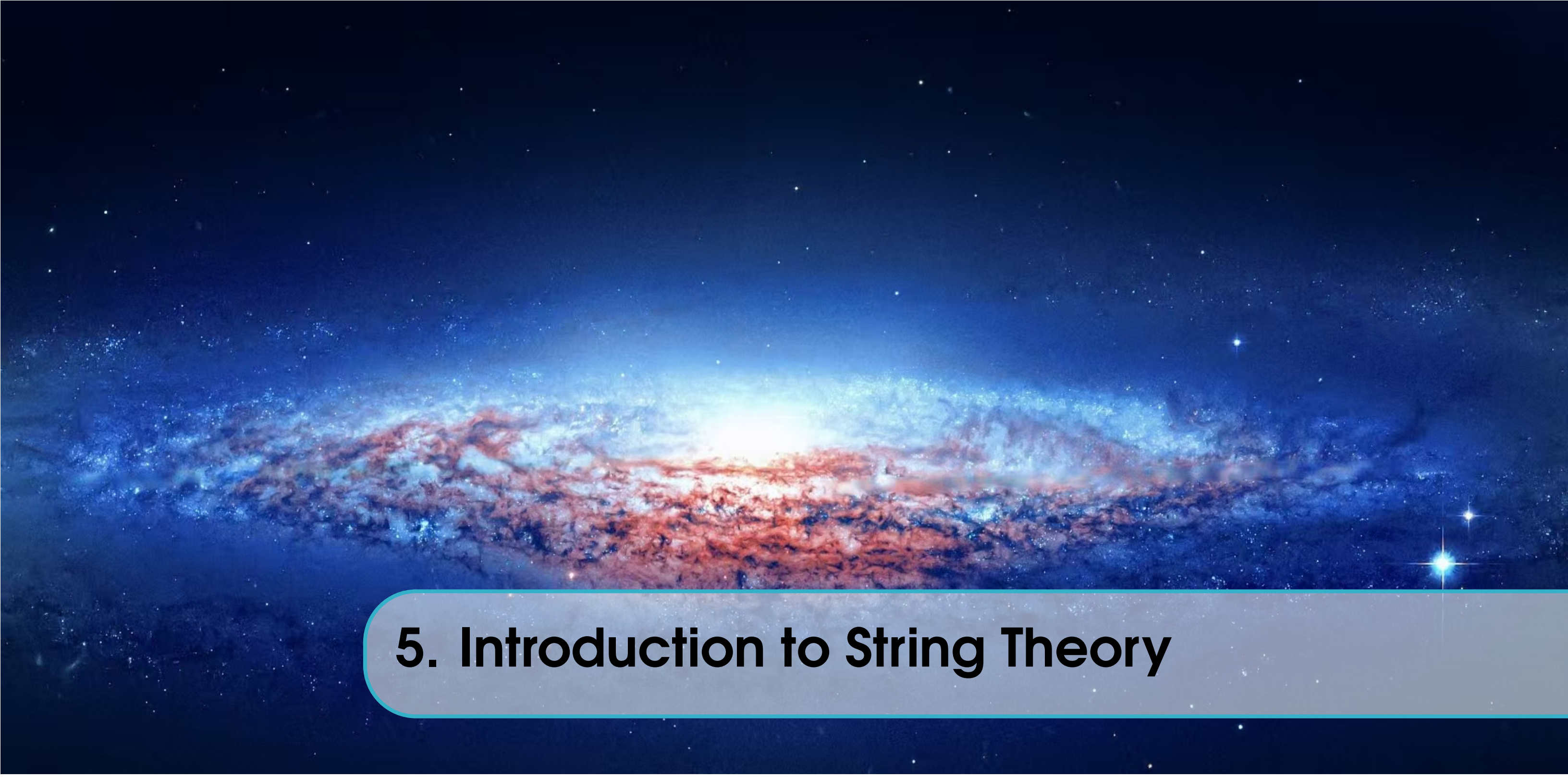
$$\tilde{\Phi} = i\sigma_2 \Phi^* = \frac{1}{\sqrt{2}}$$

All in all,

$$\mathcal{L}_{\mathrm{SM}} = -\frac{1}{4}G^a_{\mu\nu}G^{a\mu\nu} - \frac{1}{4}W^i_{\mu\nu}W^{i\mu\nu} - \frac{1}{4}B_{\mu\nu}B^{\mu\nu} + \sum_\psi \bar{\psi}i\gamma^\mu D_\mu\psi + |D_\mu\Phi|^2 - \mu^2\Phi^\dagger\Phi - \lambda(\Phi^\dagger\Phi)^2$$

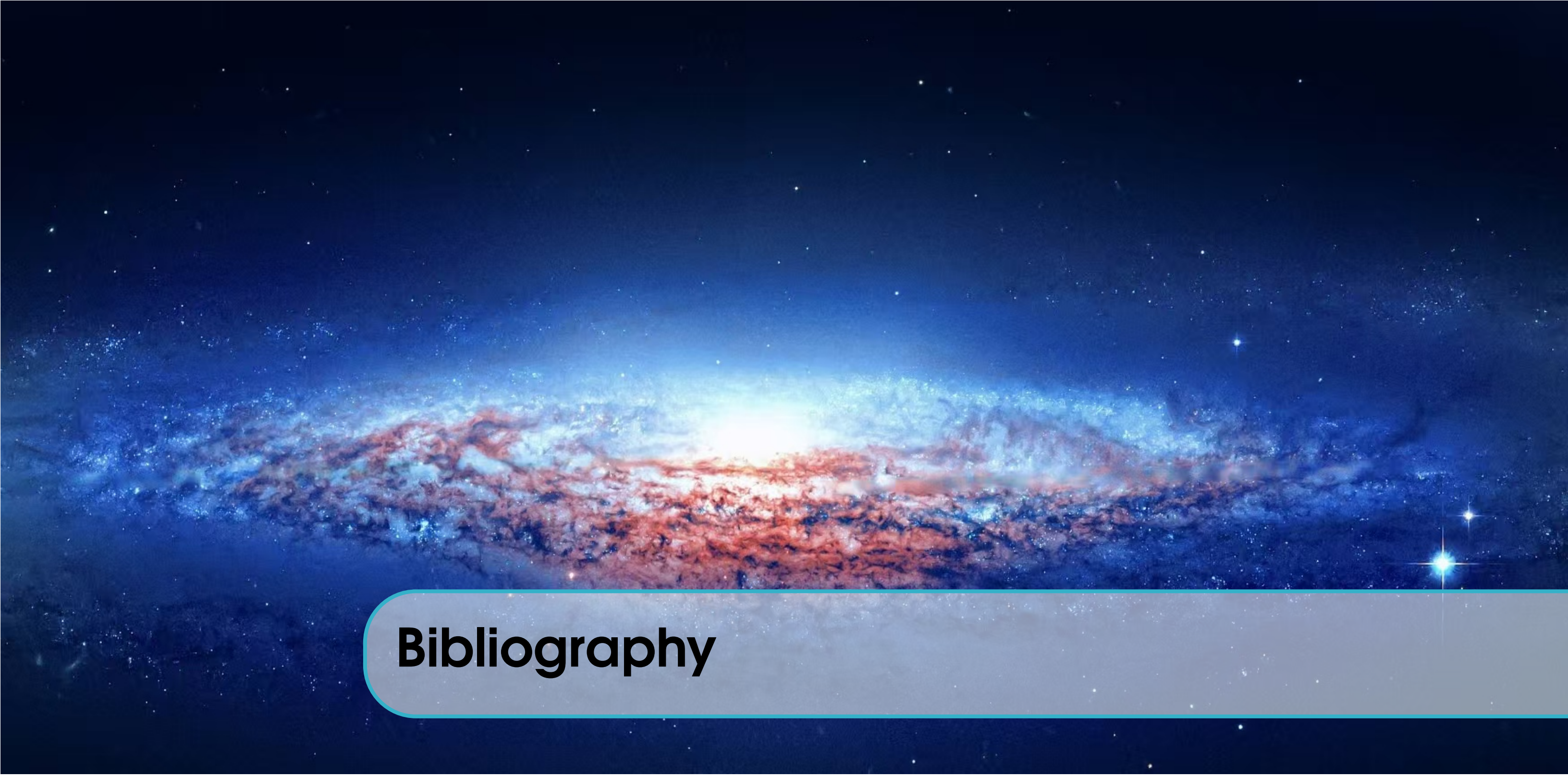
$$-\bar{Q}_L Y_u \tilde{\Phi} u_R - \bar{Q}_L Y_d \Phi d_R - \bar{L}_L Y_l \Phi l_R + \mathrm{h.c.}$$

4.2 God particle (Higgs boson)



5. Introduction to String Theory

A. Appendix and Exercise Solutions



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